

Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Inference

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Abstract

Mutual information (MI) measures association between two variables. This thesis asks whether two independently sampled populations have equal MI, even when their joint probability tables differ. The usual Normal Wald test uses a normal approximation that can be inaccurate in small or sparse samples.

The thesis develops an Expanded Welch–Satterthwaite adjustment for this comparison. It derives how changing one observation affects the estimated MI variance, including changes to the cell probabilities and their row and column totals. That calculation determines the degrees of freedom for a Student t distribution. Expanded Welch retains Wald’s estimated MI difference and standard error; it changes only the distribution used to judge the resulting statistic.

The methods are evaluated in 3,111 fixed statistical settings, each with 20,000 independently sampled table pairs. The design includes square tables from 2×2 to 8×8 , selected rectangular tables, unequal sample sizes, sparse cells, and populations with different distributions but equal MI. At a nominal 5% threshold, the experiment measures false positives, detection when MI differs, and the frequency of usable results.

Expanded Welch reduces excessive Wald false positives in some settings, but the same adjustment also reduces rejection elsewhere. When both tests return a p-value, its Student t distribution gives a p-value no smaller than Wald’s. It therefore lowers rejection where Wald already rejects too rarely and can reduce detection when MI differs. In sparse samples, Expanded Welch more often fails to return a usable p-value under the implemented rule. Wald is closer to the target 5% rate in all 24 main null settings with sample sizes 500 or 1,000.

Both methods can be affected by errors in their shared standard error. In a supplementary null comparison, replacing the sample-specific standard errors with a fixed pilot estimate brings Wald’s rejection rate closer to 5%. For the settings studied, Wald remains the simpler general baseline. Expanded Welch is a potential additional check when a separate, closely matched calibration study shows that Wald rejects equal MI too often.

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Chapter 1

Introduction

Suppose two independent groups are observed on the same categorical variables X and Y . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the population MI values differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether X and Y are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where P and Q are the two joint distributions. The samples from P and Q are independent of each other; that sampling assumption does not imply independence of X and Y within either population.

1.1 Why the comparison is difficult

MI uses the probabilities in every cell of a table and the totals of its rows and columns. In a small sample, some cells can be empty even when their population probabilities are positive. The observed tables determine both the estimated MI difference and its standard error, so sparse data can affect both. The Normal Wald test compares their ratio with a standard normal distribution, but in small samples it may reject equal MI more or less often than the intended 5%.

Expanded Welch offers a possible adjustment. It uses the same estimated difference and standard error as Wald, but compares their ratio with a Student t

distribution. Choosing that Student t distribution requires a calculation specific to MI. Its degrees of freedom depend on how much the estimated MI variance changes from sample to sample. To calculate that change, we must account for the fact that changing one cell also changes its row and column totals and the MI values calculated from them. This thesis derives that calculation and tests whether the adjustment helps.

Expanded Welch can reduce false positives where Wald rejects too often, but it can also reduce detection where Wald already rejects too rarely. In sparse samples, it may fail to return a usable p-value when Wald succeeds. The experiment therefore examines false positives, detection and the frequency of usable results together.

1.2 Research questions

The thesis asks two main questions. First, how can changes in the estimated MI variance be used to choose the degrees of freedom for a test using a Student t distribution?

Second, does the resulting Expanded Welch test offer a useful improvement over Normal Wald? The experiment checks how false positives, detection and validity change with table size, uneven cell probabilities, the amount and location of association, unequal sample sizes and larger samples. It also measures the extra computation required.

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance. The calculation includes the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this calculation to choose the degrees of freedom for a two-sample test using a Student t distribution.
3. **Empirical contribution.** A reproducible experiment measures false positives, detection, usable results and runtime on fixed populations with known MI. It covers 2×2 through 8×8 square tables, rectangular tables, unequal samples and very sparse settings.
4. **Practical contribution.** Expanded Welch can help when Wald rejects equal MI too often. It can also make an already-low rejection rate lower and return

fewer usable results in sparse samples. Changing the reference distribution alone cannot correct errors in the shared MI difference or standard error.

1.4 Scope and organisation

The analysis concerns independent samples recorded in count tables with the same row and column categories. The main derivation assumes every population cell has positive probability and that its leading large-sample MI variance is greater than zero. Exact independence is considered separately because that variance becomes zero. Within this setting, the experiment compares Normal Wald with Expanded Welch using a two-sided 5% test. It does not establish performance for paired data, continuous MI estimators or a number of categories that grows with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 presents and interprets the main comparisons in sequence. Chapter 7 draws out their broader implications and limits, and Chapter 8 states the contribution and recommendation. The appendices give the detailed algebra, population settings, simulation diagnostics and supporting figures.

Chapter 2

Background and Literature Review

Testing whether two populations have equal MI requires an estimate from each sample and a measure of its sampling error. This chapter develops those inputs and reviews methods for comparing them.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = - \sum_x p_X(x) \log p_X(x). \quad (2.1)$$

Entropy summarises the uncertainty of one random variable before its value is observed ([Shannon, 1948](#)).

For a pair (X, Y) , the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X | Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y .

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \dots, r\}$ and $Y \in \{1, \dots, c\}$. Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.2)$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.3)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.4)$$

Here the summations are nested: for each row i , the inner sum runs over all columns j . Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y) , we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.5)$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. Our null requires only this one number to agree, not that the two tables match cell by cell.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij} . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.6)$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.7)$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts vary between samples but must sum to n . An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its small-sample behaviour depends on the complete probability table. In particular, rare cells can change this behaviour even when the categories and sample size are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the plug-in estimator has a positive leading bias as the sample size grows:

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.8)$$

The leading term grows with table size and shrinks as the sample grows. It does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.9)$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

2.3 Sampling variance of plug-in MI

Comparing two MI estimates requires knowing how much each varies across repeated samples. We derive the large-sample approximation

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.10)$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeyer, 1989; Brillinger, 2004).

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (2.11)$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u \frac{1}{u} = \log u + 1. \quad (2.12)$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \quad (2.13)$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \hat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \hat{u} - u$, which is the sampling error in the estimated probability. Substituting

these quantities into the general formula gives

$$\hat{u} \log \hat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\hat{u} - u)}_{\text{first-order change}}. \quad (2.14)$$

Subtracting the population value from both sides makes the change explicit:

$$\hat{u} \log \hat{u} - u \log u \approx (\log u + 1)(\hat{u} - u). \quad (2.15)$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} (\log p_{ij} + 1)(\hat{p}_{ij} - p_{ij}) \\ &\quad - \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) \\ &\quad - \sum_j (\log p_{+j} + 1)(\hat{p}_{+j} - p_{+j}). \end{aligned} \quad (2.16)$$

The margin errors are sums of the corresponding cell errors:

$$\hat{p}_{i+} - p_{i+} = \sum_j (\hat{p}_{ij} - p_{ij}), \quad \hat{p}_{+j} - p_{+j} = \sum_i (\hat{p}_{ij} - p_{ij}). \quad (2.17)$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) &= \sum_i \left[(\log p_{i+} + 1) \left\{ \sum_j (\hat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.18)$$

The inner sum over j is the probability error for a fixed row i . It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\hat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &= \log \left(\frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &= \ell_P(i, j) - 1. \end{aligned} \quad (2.19)$$

Therefore,

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}). \quad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\hat{p}_{ij} - p_{ij}) = \sum_{i,j} \hat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \quad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}) - \sum_{i,j} (\hat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.22)$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a . An empirical cell probability is the proportion of observations in that cell, so

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \quad (2.23)$$

Substituting this expression into Equation (2.22) first gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \quad (2.24)$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^n \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}. \end{aligned} \quad (2.25)$$

The first sum inside the brackets selects the single cell containing Z_a , giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i, j) is simply that cell's pointwise MI minus the population average, $\ell_P(i, j) - I(P)$. Its mean is zero because

$$\begin{aligned} E_P\{\ell_P(X, Y) - I(P)\} &= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\} \\ &= \underbrace{\sum_{i,j} p_{ij} \ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_1 = 0. \end{aligned} \quad (2.26)$$

The centred function $\ell_P(i, j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \text{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.27)$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \text{Var}\{\hat{I}(P)\} &\approx \text{Var}\left\{\frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2} \sum_{a=1}^n \text{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2} \sum_{a=1}^n V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \quad (2.28)$$

Averaging n independent contributions therefore divides their variance by n . For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\hat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \quad (2.29)$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i, j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to Testing MI Differences

This section reviews methods for estimating uncertainty in MI and testing differences between two MI values. The question in this thesis is whether two populations have the same MI, $I(P) = I(Q)$. Their cell probabilities may differ even when their MI values are equal.

Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For two independent MI estimates, the two first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.30)$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\hat{I}(P) - \hat{I}(Q)}{\sqrt{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q}} \xrightarrow{d} N(0, 1) \quad \text{under } I(P) = I(Q). \quad (2.31)$$

Here $\hat{V}(P)$ and $\hat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can

vary between samples. The Wald test still uses a normal distribution; it makes no separate small-sample adjustment for that standard-error variability.

The Welch–Satterthwaite principle.

Welch’s test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means $\bar{X}_P, \bar{X}_Q$ and variances s_P^2, s_Q^2 , the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \quad (2.32)$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\frac{(s_P^2/n_P)^2}{n_P - 1} + \frac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \quad (2.33)$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both component values affect ν . Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

Welch compares the statistic with a Student t distribution using the calculated ν . At the same significance level, this requires a larger absolute statistic to reject the null than a standard normal distribution does. The difference between the rejection thresholds from the Student t and standard normal distributions becomes smaller as ν increases. For MI, the estimated variance contributions are $\hat{V}(P)/n_P$ and $\hat{V}(Q)/n_Q$. Their component degrees of freedom need their own calculation because $\hat{V}(P)$ and $\hat{V}(Q)$ are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example, Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples. Expanded Welch applies this approach to the estimated MI variance. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings. The MI comparison in Section 6.5 has the opposite problem: the average squared plug-in standard error exceeds the measured variance of the

estimated difference. The larger cutoff leaves that variance error unchanged and makes rejection less likely still.

Prior MI inference and related comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often each test rejects when new samples are drawn from fixed populations.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student t distribution, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.34)$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s_{H,P}^2$ and $s_{H,Q}^2$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

In a segregation application, Mora and Ruiz-Castillo (2009) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature (Mora and Ruiz-Castillo, 2011). By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k -sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap would draw many new samples from the observed tables to estimate how much the test statistic varies. This requires repeating the calculation many times (Efron, 1979). A simple permutation test instead combines the two samples and shuffles their group labels. That shuffle is justified when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Shuffling labels could therefore test

a stronger claim than this thesis asks about (Chung and Romano, 2013). Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

2.5 Previous work and this thesis’s contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson’s entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying the latter approach to MI requires calculating how its estimated variance changes with the observed table.

A new observation changes the empirical cell probabilities, row and column totals, and pointwise MI values. The remaining calculation is how these changes affect the estimated MI variance. This thesis uses that calculation to choose the degrees of freedom for a Student t distribution, then evaluates whether it improves false positives and detection. Both tests start with the same estimated MI difference and standard error. Chapter 3 defines this shared statistic before Chapter 4 derives the additional variance calculation.

Chapter 3

The Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent $r \times c$ count tables with totals n_P and n_Q . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling). *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities). *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI). *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units). *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives $\hat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\hat{\Delta}_{\text{BC}} = \left\{ \hat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \hat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \quad (3.7)$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the fixed numbers of row and column categories, even when some are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\hat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (3.8)$$

Define $\hat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.9)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.10)$$

The sign of T indicates which sample has the larger corrected MI estimate. Its magnitude measures the difference relative to its estimated standard error. Under H_0 , we compare T with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald reference and motivation for expansion

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0, 1) \quad \text{under } H_0. \quad (3.11)$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.12)$$

where Φ is the standard normal distribution function.

Why consider an expanded reference?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the variability of that variance estimate to choose a Student t distribution instead. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P , to first order,

$$\mathbb{E}\{\hat{V}(P)\} \approx V(P). \quad (3.13)$$

Chapter 4 derives $\text{Var}\{\hat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3. It changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples.

4.1 Choosing Degrees of Freedom for Expanded Welch

The Welch–Satterthwaite equation combines the two estimated MI variance contributions (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \nu_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \nu_Q}. \quad (4.1)$$

The variance contributions $\hat{V}(P)/n_P$ and $\hat{V}(Q)/n_Q$ already appear in the shared standard error. The new quantities are the component degrees of freedom ν_P and ν_Q . They describe how much each estimated MI variance varies between samples relative to its mean.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\hat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change

tells us how variable $\hat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one component.

Satterthwaite approximates the positive variance estimate $\hat{V}(P)$ by the scaled chi-squared variable

$$\mathbb{E}\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.2)$$

The scale and degrees of freedom are chosen to match the mean and variance of $\hat{V}(P)$.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathbb{E}(\chi_{\nu_P}^2) = \nu_P, \quad \text{Var}(\chi_{\nu_P}^2) = 2\nu_P. \quad (4.3)$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathbb{E}\left[\mathbb{E}\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}\right] = \mathbb{E}\{\hat{V}(P)\}. \quad (4.4)$$

Its variance is

$$\text{Var}\left[\mathbb{E}\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}\right] = \frac{[\mathbb{E}\{\hat{V}(P)\}]^2}{\nu_P^2} \text{Var}(\chi_{\nu_P}^2) \quad (4.5)$$

$$= \frac{2[\mathbb{E}\{\hat{V}(P)\}]^2}{\nu_P}. \quad (4.6)$$

Matching this quantity to $\text{Var}\{\hat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathbb{E}\{\hat{V}(P)\}]^2}{\text{Var}\{\hat{V}(P)\}}}. \quad (4.7)$$

The numerator depends on the mean of the estimated MI variance; the denominator measures how much that estimate fluctuates between samples. To first order, $E\{\widehat{V}(P)\} \approx V(P)$, so the remaining task is to calculate $\text{Var}\{\widehat{V}(P)\}$.

4.2 How the MI Variance Changes with the Table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.8)$$

where $\delta_{(x,y)}$ places probability one on cell (x, y) . At $\varepsilon = 0$, the distribution is P . Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \quad (4.9)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.10)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.11)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.12)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.13)$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.14)$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 &= \frac{\mathbf{1}\{i = x, j = y\} - p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i = x\} - p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j = y\} - p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \end{aligned} \quad (4.15)$$

The joint term supplies -1 , while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.16)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.17)$$

The first sum is $\ell_P(x, y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1 ; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.18)$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.19)$$

Write this variance in second-moment form:

$$V(P) = \text{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \quad (4.20)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.21)$$

Differentiating Equation (4.20) gives

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

It remains to differentiate

$$E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.23)$$

The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.24)$$

The first sum is

$$\ell_P(x, y)^2 - E_P \{ \ell_P(X, Y)^2 \}. \quad (4.25)$$

For the second sum, define the probability-weighted row and column means

$$E_P \{ \ell_P(X, Y) \mid X = x \} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.26)$$

$$E_P \{ \ell_P(X, Y) \mid Y = y \} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.27)$$

Substitution of Equation (4.15) gives

$$\begin{aligned} &\sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}} - \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}} + I(P) \\ &= \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P). \end{aligned} \quad (4.28)$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in

Equation (4.25) with the derivative of $-I(P)^2$:

$$\begin{aligned}
& \ell_P(x, y)^2 - E_P\{\ell_P(X, Y)^2\} - 2I(P)\{\ell_P(x, y) - I(P)\} \\
&= \ell_P(x, y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x, y) + 2I(P)^2 \\
&= \ell_P(x, y)^2 - 2I(P)\ell_P(x, y) + I(P)^2 - V(P) \\
&= \{\ell_P(x, y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $E_P\{\ell_P(X, Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.28) gives

$$\begin{aligned}
g_P(x, y) &= \{\ell_P(x, y) - I(P)\}^2 - V(P) \\
&\quad + 2\left[\ell_P(x, y) - E_P\{\ell_P(X, Y) \mid X = x\} \right. \\
&\quad \left. - E_P\{\ell_P(X, Y) \mid Y = y\} + I(P)\right].
\end{aligned} \tag{4.29}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \tag{4.30}$$

The sensitivities average to zero under P , as shown in Appendix A. Their variation, not their average, determines how much the estimated MI variance changes between samples.

Assumption 6 (Regular variance sensitivity). *For the first-order Expanded Welch construction, $\tau^2(P) > 0$ and $\tau^2(Q) > 0$.*

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P , that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\hat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\delta_{Z_a^{(P)}} - P\}. \tag{4.31}$$

The plug-in $\hat{V}(P)$ applies the variance calculation to this empirical table. Equation (4.21) gives its first-order change for one observation's direction. Because a first-order change is linear in the table error (van der Vaart, 1998, Chapter 20), the changes from the observations can be averaged:

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

To interpret Equation (4.35), divide the sampling variance of $\hat{V}(P)$ by the square of the population quantity it estimates:

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.36)$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution. Greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\hat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[\hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.37)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i,j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k,l) \right\}^2. \quad (4.38)$$

The estimated component degree of freedom is

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (4.39)$$

Repeating the same calculation for the second table gives $\hat{\nu}_Q$.

Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\hat{\nu}_{\text{expanded}} = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \hat{\nu}_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \hat{\nu}_Q}. \quad (4.40)$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.41)$$

where F_{t_ν} denotes the cumulative distribution function of a Student t variable with ν degrees of freedom.

For a fixed finite T , the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. Thus every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. The correction cannot raise rejection where Wald is already too conservative. Section A.5 examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.42)$$

Their totals are $n_P = 105$ and $n_Q = 100$. In the first table, cell $(1, 1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\hat{\ell}_P(1, 1) = \log \left\{ \frac{35/105}{(50/105)(50/105)} \right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\hat{I}(P) = 0.13693$; the same calculation for the second table gives $\hat{I}(Q) = 0.25848$ nats. Since $(3 - 1)(3 - 1) = 4$, the corrected difference is

$$\hat{\Delta}_{BC} = \left(0.13693 - \frac{4}{2(105)} \right) - \left(0.25848 - \frac{4}{2(100)} \right) = -0.12059. \quad (4.43)$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1, 1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\hat{V}(P)$. Summing over all cells gives $\hat{V}(P) = 0.27189$ and $\hat{V}(Q) = 0.43371$. Hence

$$\widehat{SE} = \left(\frac{0.27189}{105} + \frac{0.43371}{100} \right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.44)$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436 , weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

The first column has the same conditional mean because this count table is symmetric. Substituting these values into Equation (4.37) gives

$$\begin{aligned}\hat{g}_P(1, 1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\hat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\hat{\tau}^2(P) = 0.93212$ and $\hat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.45)$$

Combining them with Equation (4.40) gives $\hat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\hat{V}(P)$, $\hat{V}(Q)$, the shared standard error, and T .
4	Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$.
5	Compute $\hat{\tau}^2(P)$, $\hat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, MI is zero and its first-order change is also zero. The leading changes in MI, the pointwise-MI variance $V(P)$, and the variance sensitivity $\tau^2(P)$ can therefore be compared using a second-order expansion. Appendix A.2 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.46)$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \quad (4.47)$$

For a fixed sample size, this approximation makes the degrees of freedom smaller as population MI approaches zero. The cutoff from the Student t distribution then increases, so a larger absolute statistic is needed to reject equal MI. These approximations require the fixed-margin path, fixed table size and positive cell probabilities. Using them for samples also requires a useful first-order sampling approximation. At exact independence, quadratic sampling variation replaces the first-order term; Section A.4 derives that case.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) are exact derivatives for positive population tables. They provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.2) and the Student t distribution for T are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and the degrees-of-freedom adjustment does not account for how the two move together. Appendix A.3 gives the first-order covariance between each population's MI and variance estimates. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student t distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error. Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.5 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

Chapter 5

Experimental Design

The simulation study evaluates whether Expanded Welch improves on Normal Wald when comparing the MI of two populations. It measures false positives when the population MI values are equal, detection when they differ, and how often each method returns a valid p-value. Each experimental setting starts with two joint probability tables, P and Q , whose margins and MI values are chosen before sampling. Independent samples are then drawn from each population, and both tests are applied to the resulting count tables. Because the population MI values are known, we can assess the tests against the true difference rather than the differences that arise by chance in individual samples.

5.1 Constructing fixed joint distributions

To vary association without changing category frequencies, we construct each table with prescribed row and column margins and a known MI. A binary example shows how the margins can remain fixed while X and Y become dependent.

Constructing a non-independent 2×2 joint distribution.

To obtain one observation from P , first draw X with category probabilities $(0.7, 0.3)$. With probability approximately 20.35%, set $Y = X$; otherwise, draw Y independently with the same category probabilities. Copying X makes matching categories more likely than under independence. Both ways of choosing Y give it margins $(0.7, 0.3)$, so their mixture keeps the row and column margins fixed.

For example, cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, while an independent draw gives $Y = 1$ with probability 0.7. Combining the two possibilities gives

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} \simeq 0.53273.$$

This is the first entry of P below; the other entries follow by considering their corresponding category pairs.

Construct Q in the same way, using category probabilities $(0.8, 0.2)$ and a copying probability of approximately 26.09%. The two copying probabilities are chosen to give $I(P) = 0.02$ and $I(Q) = 0.03$ nats.

The resulting joint probability tables are

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \quad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \quad (5.1)$$

Their absolute MI difference, 0.01 nats, is the horizontal-axis value used in the results figures.

Probability transfers in larger tables.

For larger tables, begin with independence by multiplying the prescribed row and column probabilities. The main construction adds the same amount to cells $(1, 1)$ and $(2, 2)$, and removes that amount from cells $(1, 2)$ and $(2, 1)$. Each affected row and column gains and loses the same amount, so its total is unchanged; all other cells stay unchanged. This four-cell transfer follows the sparse alternative of [Berrett and Samworth \(2021, Section 3\(b\)\)](#). For the 2×2 example, this transfer produces exactly the same table as the copying mechanism above.

For each population, the amount transferred is chosen to reach its target MI while keeping every cell probability positive. Before sampling, both tables are checked for positive cells, correct margins and target MI. Appendix B gives the equivalent matrix construction, its feasible range and the numerical search used to select the amount.

The wider study uses three margin profiles. For a dimension of size k , the skewed margin used in the design is

$$m_k(\rho) = \left(\rho, \frac{1 - \rho}{k - 1}, \dots, \frac{1 - \rho}{k - 1} \right). \quad (5.2)$$

The three principal profiles are specified separately for the two populations in Table 5.1. For a rectangular $r \times c$ table, the formula is applied with $k = r$ to row

margins and $k = c$ to column margins. The margin profiles also determine whether equal MI gives identical tables. At a zero MI difference, the same-margin profiles produce $P = Q$ under this construction; the different-skew profile has $I(P) = I(Q)$ but $P \neq Q$.

Table 5.1: Row and column margins used for each population in the principal design. “Uniform” means the vector with every entry equal to $1/k$.

Profile	P : rows and columns	Q : rows and columns
Uniform	uniform	uniform
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

In the usual effect direction, $I(P) = I_0$ and $I(Q) = I_0 + |\Delta|$. The reverse-direction imbalance experiment assigns the higher target to P . The statistical convention remains $\Delta = I(P) - I(Q)$, which is negative in the usual direction. Figures use the nonnegative magnitude $|\Delta|$.

The changed-cell experiments ask whether results depend on where association sits in the table. The “rare block” moves the four changed cells from the upper-left corner to rows and columns with smaller probabilities in skewed populations. The “spread” pattern changes cells throughout the table while keeping the row and column totals fixed. Each population retains its specified margins and target MI. Comparing the patterns at these matched values tests how the placement of association affects performance.

A separate comparison changes the construction rule. The log-linear method also spreads association through the table but produces different cell probabilities from the four-cell transfer at matched margins and MI. Appendix B gives the construction formulas.

5.2 Sampling, tests and measured outcomes

For each unique configuration, 20,000 independent pairs of multinomial count tables are drawn from its fixed P and Q . Normal Wald and Expanded Welch are applied to the *same* count-table pair. Differences in their test outcomes therefore arise from the methods rather than different draws.

Rejection is measured at the same two-sided nominal threshold, $\alpha = 0.05$, throughout the statistical study. For each method, the primary plotted rejection

rate is

$$\frac{\#\{\text{sample pairs giving a valid p-value no larger than } 0.05\}}{20,000}. \quad (5.3)$$

At $|\Delta| = 0$ this estimates the false-positive rate. At $|\Delta| > 0$ it estimates the probability of detecting the specified difference at the fixed nominal threshold, often called power. Closer to 0.05 means better calibration under the null. Under an alternative, higher detection is useful only alongside acceptable null calibration and a usable test result.

An invalid calculation contributes no rejection to Equation (5.3), but it does not count as evidence for the null. The valid-result rate shows how often each test returns a usable p-value. Reporting it alongside rejection distinguishes a low rejection rate from a failure to calculate the test.

The reported 95% intervals quantify the simulation uncertainty in those rates at each fixed population pair. Intervals for the difference between methods account for applying both tests to the same sampled tables. Section B.4 gives the interval calculations and rates restricted to valid results.

The “minimum expected count” diagnostic is $\min_{ij} n_P p_{ij}$ for P , or $\min_{ij} n_Q q_{ij}$ for Q . The design imposes no minimum expected-count requirement, so empty observed cells and rows remain possible. To make low validity visible, a hollow marker warns that fewer than 90% of outputs of that method were valid; 90% is a display convention, not a validated safety threshold.

5.3 Experiment families and computation

Table 5.2 records the question answered by each family. The common population construction above applies unless the named factor changes. Exact parameter sets are in Appendix B.

The principal grid crosses square shapes $2 \times 2, 3 \times 3, 5 \times 5, 8 \times 8$ with the three margin profiles in Table 5.1. It fixes $I(P) = 0.02$ nats and uses $|\Delta| \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$ nats and equal sample sizes $n_P = n_Q \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$. The other families change one or more stated factors from this principal grid.

The design has 534 distinct population pairs (P, Q) . A pair combined with sample sizes (n_P, n_Q) defines a statistical configuration. Reusing pairs at different sample sizes gives 3,111 unique configurations, each sampled 20,000 times, for 62,220,000 table pairs. Some configurations answer more than one experimental question. The within-family counts in Table 5.2 do not sum to 3,111 because the

same configuration can occur in several families. Results from the same configuration may therefore appear in more than one comparison, using the same samples.

Table 5.2: The fixed experiment families. All statistical points use 20,000 paired replicates and the 5% threshold.

Family	Configurations (<i>shape/pattern choices</i> $\times$ <i>margin choices</i> $\times$ <i>MI/construction choices</i> $\times$ <i>sample-size pairs</i>)	Factor examined
Main	648 ($4 \times 3 \times 6 \times 9$)	Square table size, three margin profiles and equal sample size.
Baseline	648 ($4 \times 3 \times (3 \times 6) \times 3$)	Three starting MI values, each crossed with six MI differences.
Changed cells	144 ($(3 \times 2 + 2) \times 1 \times 6 \times 3$)	First and rare blocks for three shapes; spread pattern for two.
Imbalance	1,056 ($4 \times 3 \times (1 + 5 \times 2) \times 8$)	One equal-MI case and five differences in each direction; eight sample-size pairs.
Larger effects	162 ($2 \times 3 \times 9 \times 3$)	Nine MI differences up to 0.2 nats.
Rectangular	108 ($2 \times 3 \times 6 \times 3$)	Two rectangular shapes: 2×3 and 3×5 .
Construction	432 ($3 \times 2 \times (2 \times 6 \times 2) \times 3$)	Two starting MI values, six differences and two construction methods.
Extreme skew	315 ($3 \times 3 \times 5 \times 7$)	Three dominant-margin pairs up to 0.9995, including $n = 1$ stress.
Rare-cell stress	140 ($((2 \times 2) \times 1 \times 5 \times 7)$)	First or rare block in two shapes, including $n = 1$ stress.
Independence	144 ($2 \times 3 \times 6 \times 4$)	Exact $I(P) = 0$, with six MI differences.
Convergence	108 ($4 \times 3 \times 2 \times 4$ $+ 3 \times 1 \times 1 \times 4$)	Equal-MI comparisons up to $n = 50,000$, plus three rare-block controls.
Runtime	60 ($5 \times 2 \times 2 \times 3$)	Full-call timings, separate from the simulation replicates.

The runtime experiment times full method calls on pre-generated tables, excluding population construction and sampling. It varies table shape, profile, sample size and null or alternative status. Each regime uses 200 table pairs, 20 warm-up calls and five timed passes. Median call times and interquartile ranges are reported in Chapter 6.

Chapter 6

Experimental Results and Discussion

The figures plot how often each test rejects across 20,000 sampled table pairs. A point at a true MI difference of zero measures false positives; points to its right measure detection when the population MIs differ. The horizontal coordinate is fixed by the population construction, not estimated from the random samples. Every plotted rate divides valid rejections by all 20,000 replicates.

Blue solid circles show Normal Wald and magenta dashed squares show Expanded Welch. The dotted line marks the target 0.05 null rejection rate. A hollow marker warns that fewer than 90% of calculations were valid. Where a zoom is shown, the upper panels use a 0–1 rejection axis and the lower panels enlarge the 0–0.15 range.

We first ask whether lower rejection improves false-positive rates, then examine its effect on detection. Sparse-sample validity shows when a low rate instead reflects failed calculations. The final comparisons examine errors shared by both methods and whether larger samples reduce them. Additional matched figures are in [Appendix C](#).

6.1 Calibration across table size and margins

The first comparison uses a 2×2 table with uniform margins in both populations. [Figure 6.1](#) shows three equal sample sizes and asks whether the lower Expanded Welch rejection rate is closer to the target 0.05 at each size.

For these 2×2 tables, both P and Q have uniform row and column margins $(1/2, 1/2)$. From left to right, the columns use equal sample sizes $n_P = n_Q = 10, 100, 1000$. Both populations use probability transfers within the first two rows

and columns, with $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The MI difference varies over $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

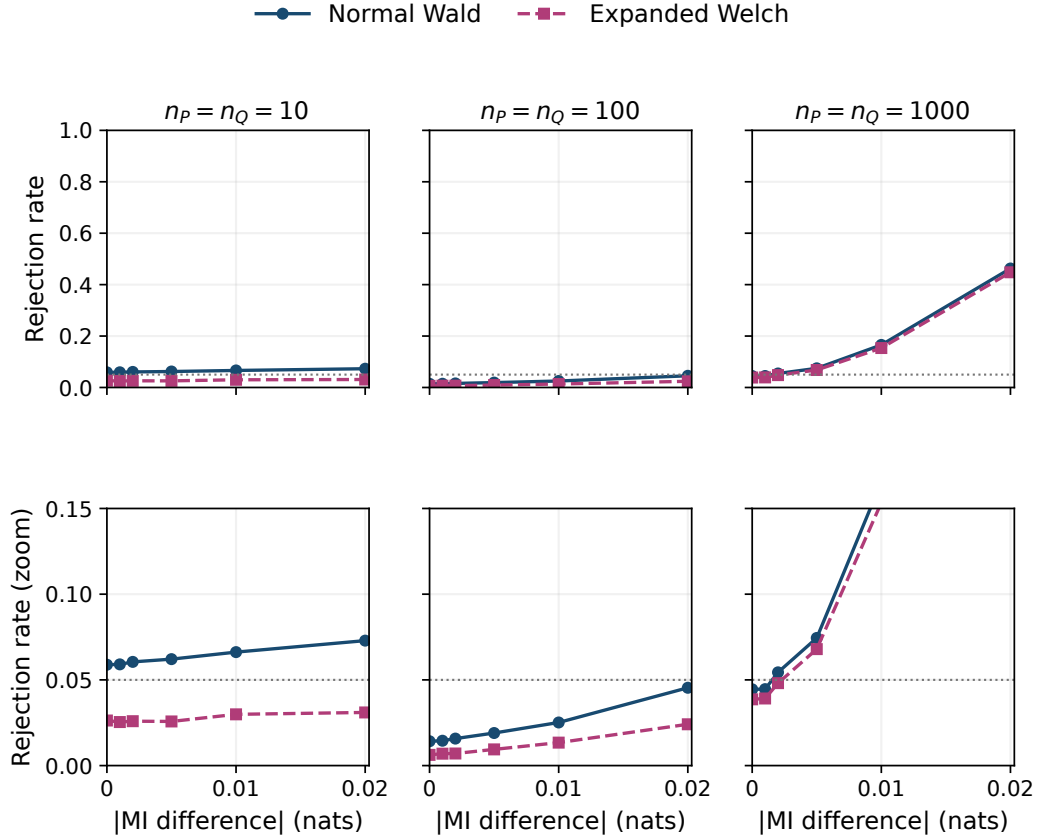


Figure 6.1: Normal Wald and Expanded Welch rejection rates for 2×2 tables under the uniform margin profile. The lower panels enlarge the 0–0.15 range.

The calibration effect depends on sample size. At zero difference, both methods reject below 0.05 at $n = 100$ and 1,000; Expanded Welch is lower still. At $n = 10$, Wald is slightly above 0.05 and Expanded Welch below it. Validity is high in all three panels, so the low null rates at the larger sizes reflect conservative rejection rather than missing outputs. Detection also changes with sample size. Rejection grows most clearly with MI difference at $n = 1,000$, where the two curves are close. At the smaller sizes, the lower Expanded Welch curve represents less detection as well as fewer false positives.

Keeping uniform margins, Figure 6.2 asks whether this sample-size trade-off changes in a larger 8×8 table.

For these 8×8 tables, both P and Q have uniform row and column margins $(1/8, \dots, 1/8)$. From left to right, the columns use equal sample sizes $n_P = n_Q = 10, 100, 1000$. Both populations use probability transfers within the first two rows

and columns, with $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The MI difference varies over $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

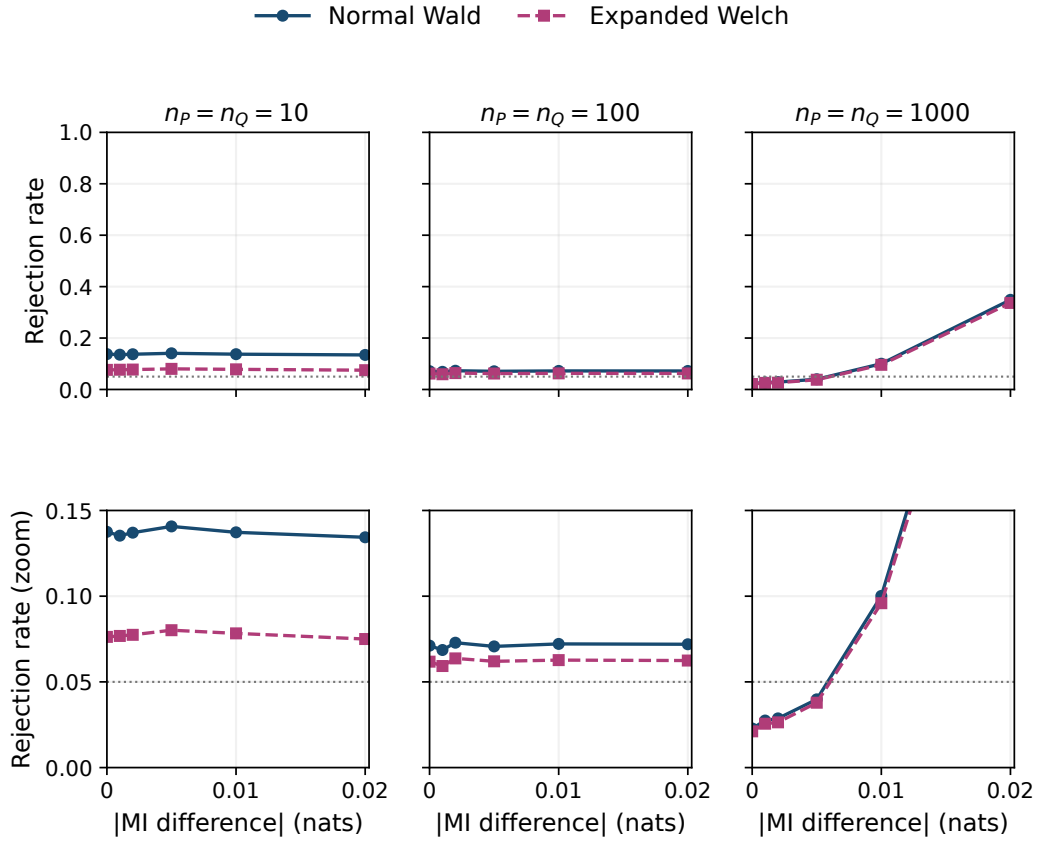


Figure 6.2: Normal Wald and Expanded Welch rejection rates for 8×8 tables under the uniform margin profile. The lower panels enlarge the 0–0.15 range.

At $n = 10$ and 100 , the zero-difference rates exceed 0.05 for both methods; Expanded Welch reduces the excess but does not always reach the target. Both methods return valid p-values throughout these panels. Unlike the smaller uniform table at $n = 100$, this setting gives the adjustment an excessive false-positive rate to reduce. Reducing that excess does not ensure strong detection. At $n = 100$, rejection changes little over the displayed alternatives; at $n = 1,000$, it grows strongly. Even with uniform margins, table size and sample size alter the trade-off. The next comparison keeps the larger table but allows the two populations' margins to differ.

For these 8×8 tables, P has row and column margins $(0.7, 0.3/7, \dots, 0.3/7)$, while Q has $(0.8, 0.2/7, \dots, 0.2/7)$. From left to right, the columns use equal sample sizes $n_P = n_Q = 10, 100, 1000$. Both populations use probability transfers within the first two rows and columns, with $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The MI difference varies over $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

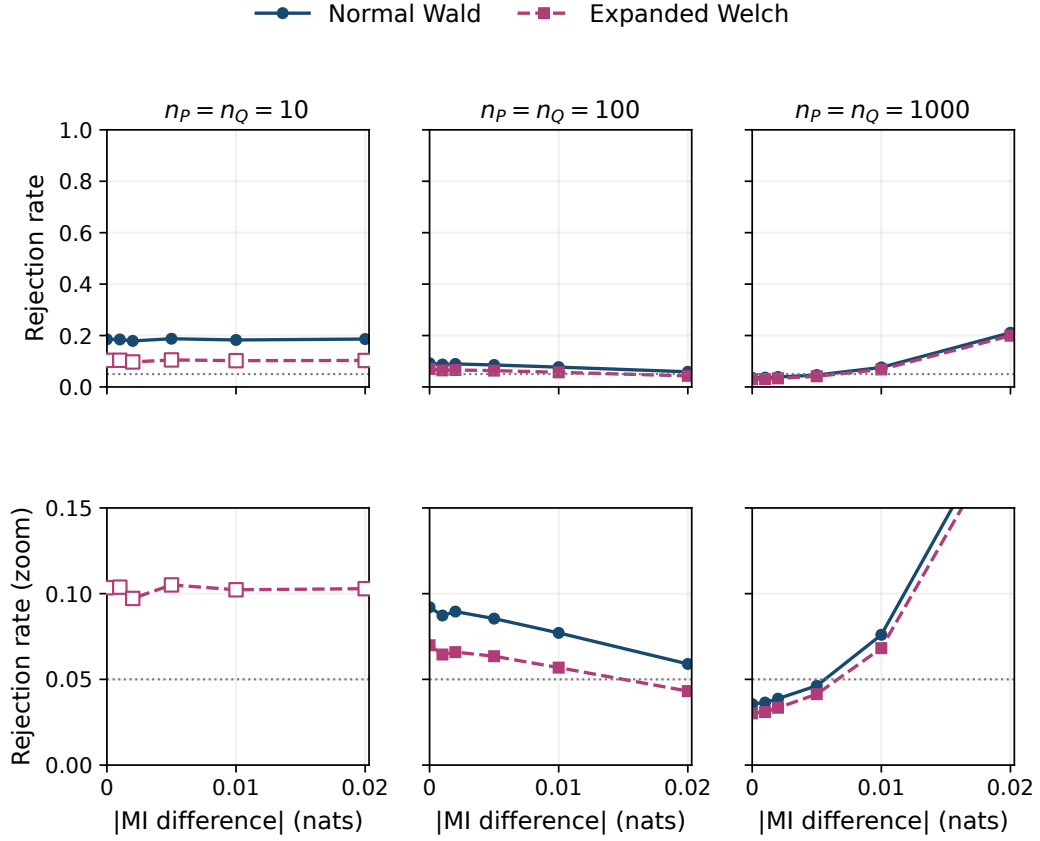


Figure 6.3: Normal Wald and Expanded Welch rejection rates for 8×8 tables under the different skew margin profile. The lower panels enlarge the 0–0.15 range.

With different margins, the adjustment still lowers an excessive null rate. At zero difference, both methods are too liberal at $n = 10$ and 100 , although Expanded Welch rejects less. Its $n = 10$ markers are hollow, so some of that reduction also reflects invalid calculations. At $n = 100$, rejection falls rather than grows over the displayed alternative range; at $n = 1,000$, detection rises but stays lower for Expanded Welch. Reducing false positives here also reduces detection.

Across the 24 main null regimes with $n = 10$ or 20 , Expanded Welch is numerically closer to 0.05 in 15, whereas Wald is closer in 9. At $n = 500$ or $1,000$, Wald is closer in all 24 main null regimes. The $n = 5$ regimes need separate treatment below because Expanded Welch never rejects there. Appendix C lists all 108 main null rates and shows selected additional curves.

Table 6.1 compares two null settings where both methods return a p-value on every sample. Brackets beside each rate give pointwise Wilson 95% intervals; the final column gives an approximate paired 95% interval for Wald minus Expanded Welch on the same samples.

Table 6.1: Rejection rates and paired differences in two fully valid null regimes.

Regime	Wald	Expanded	Wald minus Expanded
	Rate [95% interval]	Rate [95% interval]	Rate [95% interval]
8×8 , uniform, $n = 20$	0.0687 [0.0653, 0.0723]	0.0570 [0.0538, 0.0602]	0.0118 [0.0103, 0.0132]
2×2 , uniform, $n = 1000$	0.0445 [0.0417, 0.0474]	0.0387 [0.0361, 0.0414]	0.0059 [0.0048, 0.0069]

In the 8×8 row, Expanded Welch moves Wald’s above-target rate closer to 0.05 . In the 2×2 row, it moves an already-low rate farther below the target. Invalid outputs cannot explain either difference. Both methods use the same statistic, but Expanded Welch’s Student t distribution gives a p-value at least as large as Wald’s. It can therefore only lower rejection. We next examine what this means for detection when MI differs.

6.2 Detection as the true MI difference grows

The main grid shows only differences up to 0.02 nats. Figure 6.4 extends a uniform 2×2 comparison to 0.2 nats to check whether the low rejection over small effects persists as the true MI difference grows. Both populations have row and column margins $(0.5, 0.5)$, use the first 2×2 change block, and have sample sizes $n_P = n_Q = 100$. The population MI values are $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The difference grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2$ nats.

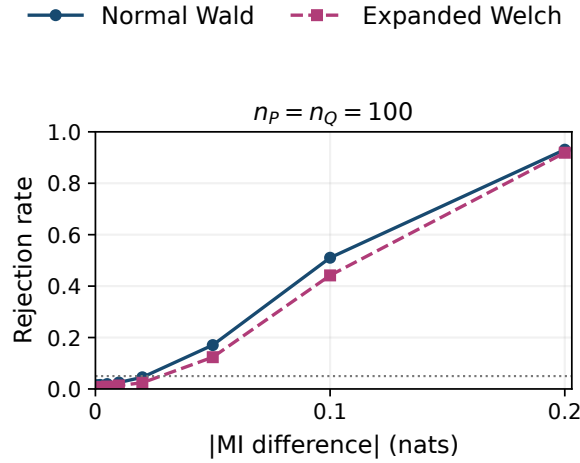


Figure 6.4: Rejection rates over wider MI differences for uniform 2×2 tables.

Both methods detect the larger MI differences much more often. At zero difference, both are below the 0.05 line; Expanded Welch is lower, and validity is high. Rejection then rises markedly as the true MI difference reaches the larger targets: by 0.2 nats both methods reject more than 90% of samples. Expanded Welch remains below Wald at intermediate differences, but the gap narrows when detection is already high.

The next question is whether the starting MI changes this trade-off. Figure 6.5 compares three starting MI values in 3×3 different-skew tables while keeping the other displayed settings fixed. From left to right, the columns use $I(P) = 0.0001, 0.001, 0.02$ nats, with $I(Q) = I(P) + |\Delta|$. The row and column margins are $(0.7, 0.15, 0.15)$ for P and $(0.8, 0.1, 0.1)$ for Q . Both populations use the first 2×2 change block and $n_P = n_Q = 100$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

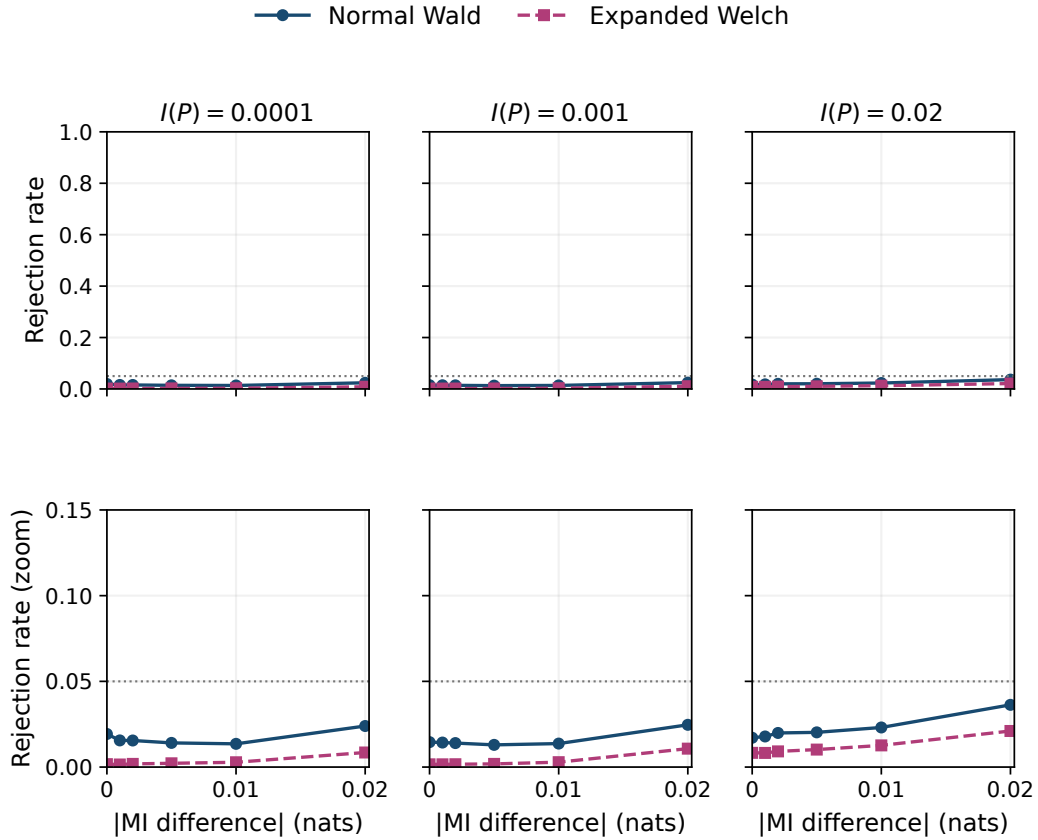


Figure 6.5: Rejection rates at three starting MI values for 3×3 tables with different margins. The lower panels enlarge the 0–0.15 range.

Starting closer to independence does not improve calibration in this comparison. For $I(P) = 0.0001, 0.001, 0.02$, the zero-difference Wald rates are 0.0193, 0.0146 and 0.0171; Expanded Welch gives 0.0017, 0.0016 and 0.0083. All these null calculations are valid. Rejection rises only modestly across the displayed alternatives, and both methods remain mostly below 0.05. Expanded Welch’s higher cutoff lowers an already-conservative null rate and further reduces detection.

The local calculation in Chapter 4 explains why the adjustment can be particularly restrictive near independence. For a fixed sample size, smaller population MI can give fewer degrees of freedom and a higher cutoff from the Student t distribution. The test then needs a larger absolute statistic to reject, while the first-order MI approximation is also less reliable. These detection rates use the common nominal 5% threshold. The methods have different false-positive rates, so the curves compare detection under the stated tests, not at matched false-positive rates.

6.3 Validity under sparse sampling

An observed table can omit many cells even when every population cell has positive probability. Figure 6.6 pushes the margins toward one dominant category while holding $n_P = n_Q = 100$. It asks whether near-zero rejection in very sparse tables means good calibration or simply a failure to calculate a test. From left to right, the dominant row and column probabilities for (P, Q) are $(0.9, 0.95)$, $(0.99, 0.995)$, and $(0.999, 0.9995)$. Each remaining margin entry divides the leftover probability equally between the two other categories. Both populations use the first 2×2 change block, with $I(P) = 0.00001$ and $I(Q) = I(P) + |\Delta|$. The difference grid is $|\Delta| = 0, 0.00001, 0.00005, 0.0001, 0.0002$ nats.

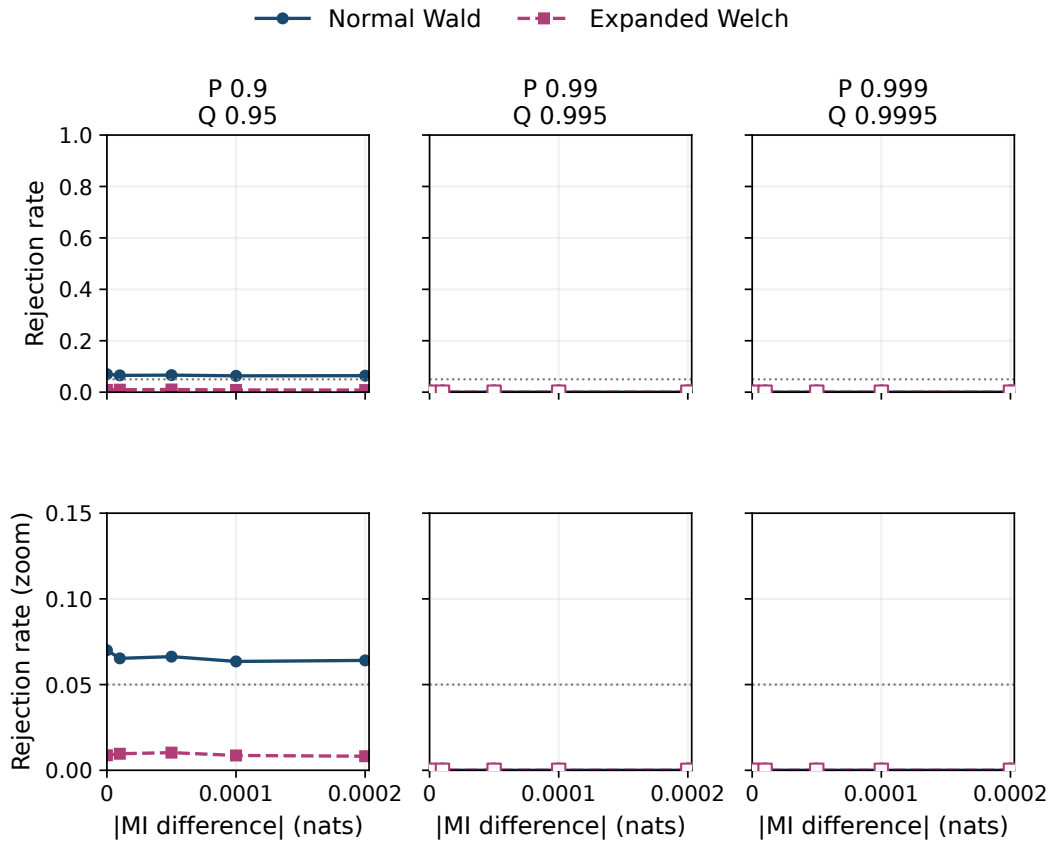


Figure 6.6: Rejection rates for 3×3 tables with increasingly dominant categories. The lower panels enlarge the 0–0.15 range.

The adjustment can reduce false positives too far even before the most extreme sparsity is reached. For dominant margins 0.9 and 0.95, the null rates are 0.0701 for Wald and 0.0088 for Expanded Welch: the adjustment moves an excessive rate past 0.05. At the most extreme margins, zero rejection mostly reflects failed calculations. With dominant probabilities 0.999 and 0.9995, both rates are zero, but only 0.0118

of Wald and 0.00005 of Expanded Welch null calculations are valid. Rejection does not meaningfully grow over the displayed alternatives there. The hollow markers show why near-zero unconditional rejection cannot be read as good calibration in sparse data.

Low validity explains much of the zero rejection in the most extreme panel. There is a second, distinct small-sample effect: Expanded Welch rejects in none of the 12 main-grid null regimes with $n = 5$, including uniform 3×3 , 5×5 and 8×8 regimes where most calculations are valid. Thus its zero rejection cannot always be explained by missing outputs. In one 8×8 different-skew alternative at $n = 5$, Wald rejects 0.30865 and Expanded Welch rejects 0.00000, while their valid-result rates are 0.83840 and 0.31415.

Wald needs a positive usable total estimated variance. Expanded Welch also needs positive finite degrees of freedom for both components and their combination. Some count tables therefore produce a Wald p-value but no Expanded Welch p-value. The primary rate counts such failures as non-rejections. The exact numerical validity criteria are given in Appendix B.5.

6.4 Checks on population and sampling choices

Margins and MI alone do not determine finite-sample performance. Moving association between the first block, rare cells and a spread pattern changes rejection rates even though each population's margins and MI stay fixed (Figure C.3). Changing from a four-cell transfer to an ordinal log-linear construction also changes rates under matched margins and MI (Figure C.7). The cell probabilities therefore matter beyond these summaries. The corresponding plots and specifications are in Appendix C.

Sampling allocation also changes the detection comparison. Reversing which population has the larger sample or higher MI changes the alternative curves (Figures C.4 and C.5). The direction of the MI difference therefore matters alongside the number of observations assigned to each population. The low null rates are not confined to square tables. Rectangular tables also show conservative rejection (Figure C.6). Calibration and validity therefore depend on the particular populations and sampling allocation.

At exact independence, the first-order MI variance is zero. Figure C.8 reports this boundary check. Section A.4 explains why a one-table independence test has a different reference distribution.

6.5 Limits of changing the reference distribution

Changing the reference distribution leaves errors in the shared statistic untouched. Both methods use the same estimated MI difference and the same plug-in standard error, so any error in either quantity affects both tests. Different small-sample biases can shift the estimated MI difference away from zero even when the population MIs are equal. At equal sample sizes, the leading bias corrections cancel in the difference. The remaining biases need not cancel when P and Q have different margins. The study does not separately measure how much this residual bias contributes to rejection.

A supplementary check shows that replacing the shared standard error can bring Wald’s false-positive rate closer to 0.05. For a 3×3 different-skew null with $I(P) = I(Q) = 0.02$ and $n = 100$, Wald rejects 0.0166 of independently simulated table pairs. Keeping the same estimated MI differences but dividing each by one fixed standard deviation estimated in a separate pilot simulation raises rejection to 0.0551. The MI differences and normal rejection cutoff are unchanged, so the increase comes from replacing the standard error. The size of the plug-in standard error helps explain this increase. In this setting, the average squared plug-in standard error is about 17% larger than the measured variance of the MI difference. A larger standard error tends to make the absolute test statistic smaller and rejection less likely.

This replacement changes two properties at once: the size of the standard error and its dependence on the estimated MI difference. A fixed standard error cannot rise or fall with the difference across samples, so the check does not separate these two effects. Appendix A.3 gives the first-order covariance between each population’s MI and variance estimates.

6.6 Larger samples and computational cost

Figure 6.7 keeps each population pair fixed under equal MI and increases the sample size. This tests whether the poor null rates are temporary small-sample effects for the chosen populations, and whether the answer changes with their common MI. The rows compare 2×2 and 8×8 tables, and the columns use common population MI values of 0.0001 and 0.02 nats. For each shape, P ’s row and column margins are $m_k(0.7)$ and Q ’s are $m_k(0.8)$, with $k = 2$ or 8 . Both populations use the first 2×2 change block. The equal sample sizes are $n_P = n_Q = 1,000, 2,500, 10,000, 50,000$, plotted on a labelled logarithmic scale. Every point has equal population MI.

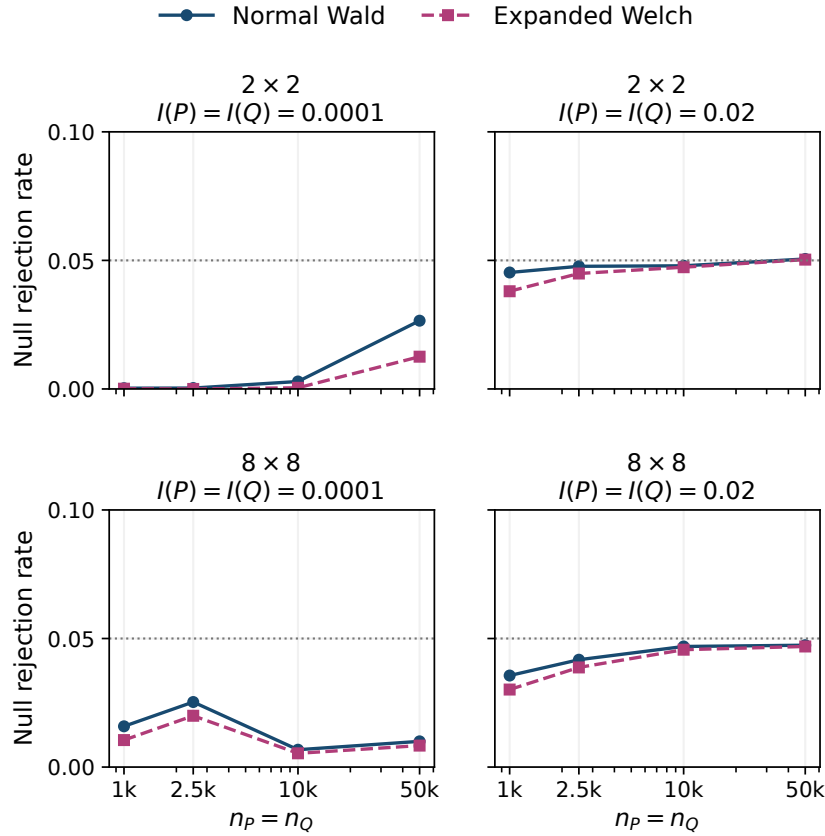


Figure 6.7: False-positive rates as sample size increases for 2×2 and 8×8 tables. The vertical axis ranges from 0 to 0.10.

In the displayed 2×2 and 8×8 settings, larger samples improve calibration much more clearly at MI 0.02 than at MI 0.0001. All displayed calculations are valid. Across all 12 first-block shape–margin combinations at $n = 50,000$, the 0.02-MI null rates range from 0.0470 to 0.0527 for Wald and 0.0466 to 0.0526 for Expanded Welch. At MI 0.0001, the ranges remain 0.0089–0.0280 and 0.0076–0.0138. Convergence can therefore be slow near independence. The sample size needed for acceptable calibration therefore depends on the population setting.

A full Expanded Welch method call has the same $O(rc)$ asymptotic complexity as Wald, but its extra sensitivity and degree-of-freedom calculations take longer in the measured implementation. Table 6.2 shows representative timings from the 60 timing regimes. These examples use different-skew null populations with $n_P = n_Q = 100$ and $I(P) = I(Q) = 0.02$. Entries give the median and interquartile range, in microseconds, over five passes on 200 pre-generated table pairs per regime. The inputs are already available when timing begins.

Table 6.2: Median method-call times and interquartile ranges, in microseconds.

Shape	Normal Wald	Expanded Welch	Expanded/Wald
2×2	85.8 [85.5, 86.3]	150.0 [149.4, 151.3]	1.75
3×3	85.1 [84.6, 85.6]	150.0 [148.9, 150.9]	1.76
3×5	85.2 [84.9, 85.5]	150.1 [149.5, 150.9]	1.76
5×5	85.3 [85.1, 85.7]	149.2 [148.5, 150.1]	1.75
8×8	88.0 [87.1, 88.9]	153.4 [152.4, 154.8]	1.74

Across the 60 timing regimes, the median call time is about 85 microseconds for Wald and 150 for Expanded Welch, a ratio of about 1.75. This extra cost is modest compared with the differences in calibration, detection and validity. Expanded Welch can help where Wald rejects equal MI too often, but its lower detection and validity elsewhere make it unsuitable as a general replacement.

Chapter 7

Implications and Limitations

7.1 Implications for using Expanded Welch

Normal Wald and Expanded Welch use the same estimated MI difference and standard error. Expanded Welch can only make a valid two-sided p-value larger. This is helpful when Wald rejects an equal-MI null too often; it is harmful when Wald already rejects too rarely. Lower rejection is therefore not by itself evidence of better calibration. In sparse samples, Expanded Welch’s extra component requirements can prevent it from returning a p-value. This also lowers the reported rejection rate, so calibration must be read alongside the frequency of usable results.

The trade-off varies with table size, margins, starting MI and sampling allocation, including when equal-MI populations have different margins. Neither method is well calibrated across all tested settings. For applications resembling a tested regime, Normal Wald remains the simpler baseline. Expanded Welch can be a supplementary check if a separate calibration study shows that Wald rejects equal MI too often and the expanded calculation is usually valid. Method choice should be set before examining the p-values.

The extra computation is not the principal obstacle. In the measured implementation, an Expanded Welch call takes about 1.75 times as long as a Wald call,

but both have the same $O(rc)$ operation growth for an $r \times c$ table. The statistical limitations matter more than this timing difference.

7.2 Limits of the evidence

The rejection rates and Monte Carlo intervals apply to the constructed population pairs. The intervals quantify uncertainty from the finite number of simulated samples; they do not describe how performance varies across untested populations.

One specific gap concerns the association patterns in equal-MI pairs. The main construction changes four cells. Rare-cell, spread and log-linear constructions test other patterns, but the study does not pair equal-MI populations whose association is placed in different cells. Such pairs would test whether the method comparison changes when the populations differ in association location as well as their margins.

The methods are studied with two independent samples from fixed categorical alphabets, a two-sided 5% threshold and the stated sample ladders. The analysis does not establish performance for paired observations, continuous MI estimators, growing alphabets or a real-data workflow. Exact independence also needs a different mathematical approximation. The first-order MI variance vanishes there. Section [A.4](#) derives the second-order approximation needed there.

Chapter 8

Conclusion

This thesis derives a Welch–Satterthwaite test for comparing MI between two independently sampled categorical populations. Its mathematical contribution is the calculation of how changing one observation affects the estimated MI variance, including changes to cell probabilities and their row and column totals. This determines the test’s degrees of freedom. The empirical contribution is a controlled evaluation of that calculation against Normal Wald. The experiment uses fixed populations whose true MI differences are known, reporting false positives, detection, validity and runtime together.

Expanded Welch reduces rejection relative to Wald and can move an excessive false-positive rate toward 5%. The same change makes an already-low rate lower and often reduces detection when MI differs. In sparse samples, it can also fail to return a p-value more often than Wald. These results show that lowering rejection is useful only when it improves calibration without an unacceptable loss of detection or validity. Changing the reference distribution leaves the estimated MI difference and standard error unchanged. Errors in those shared quantities can therefore affect both tests despite the degrees-of-freedom adjustment. Normal Wald remains the simpler general baseline for the tested task; Expanded Welch is best treated as a sensitivity check in a regime where its calibration and validity have been examined separately.

The recommendation is based on selected table sizes, margins, association patterns and sample sizes. The derivation assumes independent categorical samples with positive population cell probabilities and positive first-order MI variance; exact independence requires a different approximation. Applying the method in a new population setting requires a calibration check for that setting.

The most useful next step is to study how the estimated MI difference and its standard error move together across samples, especially near independence and in sparse tables. This could guide a comparison that accounts for errors in the statistic itself, rather than only changing its reference distribution. A likelihood-ratio or resampling method would need validation for equal population MI, including cases where the complete tables differ. A real-data application and equal-MI pairs with association in different cells would test how far the present simulation conclusions extend.

Appendix A

Supporting Calculations

These calculations support the variance sensitivity, near-independence interpretation and validity conditions in Chapter 4.

A.1 Mean of the variance influence function

The observation-level sensitivities $g_P(x, y)$ have a probability-weighted mean of zero. Averaging the one-cell perturbation directions leaves the population table unchanged:

$$\sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\} = P - P = 0. \quad (\text{A.1})$$

This weighted cancellation also holds after exact plug-in substitution. The first derivative is linear in the direction of change. Therefore, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$E_P\{g_P(X, Y)\} = \sum_{x,y} p_{xy} \left. \frac{d}{d\varepsilon} V\{(1 - \varepsilon)P + \varepsilon\delta_{(x,y)}\} \right|_0 \quad (\text{A.2})$$

$$= \left. \frac{d}{d\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\}\right\} \right|_0 \quad (\text{A.3})$$

$$= \left. \frac{d}{d\varepsilon} V(P) \right|_0 = 0. \quad (\text{A.4})$$

A.2 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t , all cells remain positive and the margins remain a and b . We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \quad (\text{A.5})$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i, j) = \log \left(\frac{q_{ij} + tH_{ij}}{q_{ij}} \right) = \log \left(1 + \frac{tH_{ij}}{q_{ij}} \right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2 H_{ij}^2}{2q_{ij}^2} + O(t^3), \quad (\text{A.6})$$

using $\log(1 + z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned} (q_{ij} + tH_{ij})\ell_{P_t}(i, j) &= (q_{ij} + tH_{ij}) \left\{ \frac{tH_{ij}}{q_{ij}} - \frac{t^2 H_{ij}^2}{2q_{ij}^2} + O(t^3) \right\} \\ &= tH_{ij} - \frac{t^2 H_{ij}^2}{2q_{ij}} + \frac{t^2 H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2 H_{ij}^2}{2q_{ij}} + O(t^3). \end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (\text{A.7})$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i, j)^2 = \left\{ \frac{tH_{ij}}{q_{ij}} + O(t^2) \right\}^2 = \frac{t^2 H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned} E_{P_t}\{\ell_{P_t}(X, Y)^2\} &= \sum_{i,j} (q_{ij} + tH_{ij}) \left\{ \frac{t^2 H_{ij}^2}{q_{ij}^2} + O(t^3) \right\} \\ &= t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (\text{A.8})$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned} \mathbb{E}_{P_t}\{\ell_{P_t}(X, Y) \mid X = i\} &= \frac{1}{a_i} \sum_j (q_{ij} + tH_{ij}) \left\{ \frac{tH_{ij}}{q_{ij}} + O(t^2) \right\} \\ &= \frac{t}{a_i} \sum_j H_{ij} + O(t^2) = O(t^2). \end{aligned}$$

The last equality uses the zero row sum of H . The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.29), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2 . Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.29) supplies the first-order term:

$$g_{P_t}(i, j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \quad (\text{A.9})$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned} \tau^2(P_t) &= \sum_{i,j} (q_{ij} + tH_{ij}) \left\{ \frac{2tH_{ij}}{q_{ij}} + O(t^2) \right\}^2 \\ &= 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \end{aligned} \quad (\text{A.10})$$

Comparing these three leading terms gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Section A.4 treats that case separately.

A.3 Dependence between the MI and variance estimates

Both $\hat{I}(P)$ and $\hat{V}(P)$ are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\text{Cov}\{\hat{I}(P), \hat{V}(P)\} \approx \frac{1}{n_P} \mathbb{E}_P[\{\ell_P(X, Y) - I(P)\}g_P(X, Y)]. \quad (\text{A.11})$$

The corresponding expression for Q replaces P and n_P with Q and n_Q . To see its direction near independence, use the fixed-margin path $P_t = (q_{ij} + tH_{ij})$ from Section A.2. For small positive t , the expansions there give

$$\ell_{P_t}(i, j) - I(P_t) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \quad g_{P_t}(i, j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \quad (\text{A.12})$$

Substituting both expressions into the first-order covariance above gives

$$\text{Cov}\{\hat{I}(P_t), \hat{V}(P_t)\} \approx \frac{2t^2}{n_P} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O\left(\frac{t^3}{n_P}\right) \quad (\text{A.13})$$

$$= \frac{2V(P_t)}{n_P} + O\left(\frac{t^3}{n_P}\right). \quad (\text{A.14})$$

The first-order covariance is therefore positive for small positive MI along this path: samples with a larger MI estimate also tend to have a larger variance estimate. The two-sample rejection rate depends on both populations and the ratio of the difference to its standard error. At exact independence, $V(P_0) = 0$ and a second-order sampling approximation is needed.

A.4 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i, j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \quad f(\varepsilon) = I(P_\varepsilon), \quad (\text{A.15})$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \quad (\text{A.16})$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i, j) - 1\} h_{ij} = - \sum_{i,j} h_{ij} = 0. \quad (\text{A.17})$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $d^2(u \log u)/du^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \quad (\text{A.18})$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \quad (\text{A.19})$$

The first sum measures squared cell changes. The last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \hat{P} - P$. Cell-probability errors typically have size proportional to $n^{-1/2}$. Squaring them makes $\hat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\hat{I}(P) \xrightarrow{d} \chi_{(r-1)(c-1)}^2 \quad (\text{A.20})$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G , test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

A.5 The combined formula when a component is zero

For positive component degrees of freedom, substituting Equation (4.39) and its Q analogue into Equation (4.40) gives the equivalent combined form

$$\hat{\nu}_{\text{expanded}} = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\hat{\tau}^2(P)/(2n_P^3) + \hat{\tau}^2(Q)/(2n_Q^3)}. \quad (\text{A.21})$$

This direct form remains defined when one table has $\hat{V} = \hat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. The implementation nevertheless requires both components to have positive, finite degrees of freedom, so it records this case as invalid.

Numerical derivative checks.

The analytic derivatives in Equations (4.18) and (4.29) were also checked by centred finite differences along the one-cell perturbation paths in Chapter 4. Small positive and negative steps were chosen to keep every cell positive; the change in MI or variance divided by the distance between the two steps agreed with its analytic derivative.

Appendix B

Experimental Details

This appendix specifies the population construction, exact parameter grids, uncertainty intervals and validity rules used in the experiments.

B.1 Additive construction

For prescribed positive row probabilities $a = (a_1, \dots, a_r)$ and column probabilities $b = (b_1, \dots, b_c)$, form the independence table $B = ab^\top$, whose cell probabilities are $B_{ij} = a_i b_j$. Let H specify which cells gain or lose probability and $t \geq 0$ control the amount transferred. Define

$$R(t) = ab^\top + tH, \quad H\mathbf{1}_c = 0, \quad \mathbf{1}_r^\top H = 0. \quad (\text{B.1})$$

Here $\mathbf{1}_k$ is a vector of k ones. The zero row and column sums of H ensure that the transfers leave both margins unchanged. The principal first-cell direction is zero outside rows and columns 1 and 2, where it has the block

$$H_{1:2,1:2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (\text{B.2})$$

The rare-cell direction places the same block in the last two rows and columns. The spread direction is

$$H_{ij} = u_i v_j, \quad u_i = -1 + \frac{2(i-1)}{r-1}, \quad v_j = -1 + \frac{2(j-1)}{c-1}. \quad (\text{B.3})$$

Both score vectors sum to zero, so every direction preserves margins.

The largest nonnegative strength before a decreasing cell reaches zero is

$$t_{\max} = \min_{H_{ij} < 0} \frac{B_{ij}}{-H_{ij}}. \quad (\text{B.4})$$

Evaluate the path endpoint MI. A target MI must lie strictly below this endpoint, within the stated numerical tolerance, so the chosen strength satisfies $0 \leq t < t_{\max}$. This endpoint is specific to the chosen transfer pattern, not the largest possible MI over all tables with those margins. For a positive target, Brent's one-dimensional root search (Brent, 1973) solves $I\{R(t)\} = I_{\text{target}}$ on $[0, t_{\max}]$; a zero target uses $t = 0$. The final table is accepted only if every cell is strictly positive, the sum is one, both margins agree with their prescribed vectors, and achieved MI agrees with the target. The normalisation and margin tolerances are 10^{-12} ; the MI tolerance is $10^{-12} + 10^{-8}I_{\text{target}}$.

Equivalence to the binary sampling mechanism.

In the binary example, both variables have category probabilities $a = (a_1, a_2)$. With probability w , Y copies X , giving joint probabilities a_1, a_2 on the diagonal and zero elsewhere. With probability $1 - w$, Y is drawn independently, giving $B = aa^\top$. Combining these two possibilities gives

$$R = (1 - w) \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + w \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix} \quad (\text{B.5})$$

$$= \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + w a_1 a_2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (\text{B.6})$$

The second line uses $a_1 + a_2 = 1$, so $a_1 - a_1^2 = a_1 a_2$ and $a_2 - a_2^2 = a_1 a_2$. Thus the copying mechanism is exactly the first-cell additive construction with $t = w a_1 a_2$. For the margins in Chapter 5, $t_P \simeq 0.04272945$ and $t_Q \simeq 0.04174587$ give $w_P \simeq 0.20347356$ and $w_Q \simeq 0.26091171$, respectively.

Row and column margins follow Table 5.1, with its formula applied separately to the two dimensions. The extreme regimes specify their dominant probabilities directly in Table B.1.

B.2 Log-linear comparison

The log-linear comparison uses the scores u_i, v_j from Equation (B.3), but starts from positive cell weights

$$K_{ij}(\lambda) = \exp(\lambda u_i v_j) \quad (\text{B.7})$$

instead of adding tH to the independence table B . Iterative proportional fitting (Deming and Stephan, 1940) scales rows and columns of these weights until the prescribed margins are restored. At $\lambda = 0$, this gives the independence table; a numerical search then chooses λ to reach the target MI. The final table passes the same positivity, normalisation, margin and target-MI checks as the additive table. Unlike the first-block additive construction used in the comparison, it changes cells throughout the table. Its cell probabilities can therefore differ from additive tables with the same margins and MI.

B.3 Statistical experiment grids

Unless a row of Table B.1 says otherwise, both populations use the first-cell additive direction, $I(P) = I_0$, $I(Q) = I_0 + |\Delta|$, and the three standard margin profiles. The symbol n means $n_P = n_Q = n$. Only targets producing strictly positive population tables are used. At $n = 1$, neither statistic is evaluable, so all sampled results are recorded as invalid.

Table B.1: Exact statistical experiment grids; MI values are in nats.

Family	Settings
Main	Shapes $2 \times 2, 3 \times 3, 5 \times 5, 8 \times 8$; three profiles; $I_0 = 0.02$; $ \Delta \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$; $n \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$.
Baseline	Main shapes and profiles; $I_0 \in \{0.0001, 0.001, 0.02\}$; main difference grid; $n \in \{20, 100, 1000\}$.
Changed cells	Shapes $3 \times 3, 5 \times 5, 8 \times 8$; different-skew margins; $I_0 = 0.0001$; $ \Delta \in \{0, 0.0001, 0.00025, 0.0005, 0.001, 0.002\}$; $n \in \{20, 100, 1000\}$. First and rare blocks for all shapes; spread pattern for 5×5 and 8×8 .
Imbalance	Main shapes, profiles and difference grid; $I_0 = 0.02$. Sample pairs are $(50, 50), (50, 100), (100, 50), (50, 250), (250, 50), (50, 500), (500, 50), (500, 500)$. Both Q -higher and P -higher MI directions.
Larger effects	Shapes $2 \times 2, 3 \times 3$; three profiles; $I_0 = 0.02$; main difference grid plus $0.05, 0.1, 0.2$; $n \in \{20, 100, 1000\}$.
Rectangular	Shapes $2 \times 3, 3 \times 5$; three profiles; $I_0 = 0.02$; main difference grid; $n \in \{20, 100, 1000\}$.

Family	Settings
Construction	Shapes $3 \times 3, 5 \times 5, 8 \times 8$; uniform and different-skew profiles; $I_0 \in \{0.0001, 0.02\}$; main difference grid; $n \in \{20, 100, 1000\}$; additive first block and ordinal log-linear constructors.
Extreme skew	Shapes $2 \times 2, 3 \times 3, 8 \times 8$; dominant margin probability pairs $(\rho_P, \rho_Q) \in \{(0.9, 0.95), (0.99, 0.995), (0.999, 0.9995)\}$; $I_0 = 0.00001$; $ \Delta \in \{0, 0.00001, 0.00005, 0.0001, 0.0002\}$; $n \in \{1, 2, 5, 20, 100, 1000, 10000\}$.
Rare-cell stress	Shapes $3 \times 3, 8 \times 8$; dominant pair $(0.9, 0.95)$; first and rare blocks; $I_0 = 0.00001$; $ \Delta \in \{0, 0.00001, 0.000025, 0.00005, 0.0001\}$; same n ladder as extreme skew.
Independence	Shapes $2 \times 2, 8 \times 8$; three profiles; $I_0 = 0$; main difference grid; $n \in \{2, 20, 100, 1000\}$.
Convergence	Main square shapes and profiles; equal MI $I(P) = I(Q) \in \{0.0001, 0.02\}$; $n \in \{1000, 2500, 10000, 50000\}$. Additional rare-block controls use $3 \times 3, 5 \times 5, 8 \times 8$, different-skew margins and common MI 0.0001.
Runtime	Shapes $2 \times 2, 3 \times 3, 3 \times 5, 5 \times 5, 8 \times 8$; uniform and different-skew profiles; $I_0 = 0.02$; $ \Delta \in \{0, 0.02\}$; $n \in \{20, 100, 1000\}$. Each of the 60 regimes uses 200 pre-generated table pairs, 20 warm-up calls and five timed passes.

B.4 Rejection rates and simulation uncertainty

The plotted rejection rate divides valid rejections by all 20,000 sampled table pairs. The method-valid conditional rate instead divides by that method's valid outputs; it is undefined if there are none. The common-valid rate restricts the calculation to pairs for which both methods are valid. Comparing unconditional and common-valid rates separates the effect of the reference distribution from lost validity.

For the paired comparison, let D_s equal the Wald valid-rejection indicator minus the Expanded Welch valid-rejection indicator for sampled pair s , matching the sign in Table 6.1. Its average is $\bar{D} = (1/20,000) \sum_{s=1}^{20,000} D_s$, with approximate 95% interval

$$\bar{D} \pm 1.96 \left\{ \frac{1}{20,000(19,999)} \sum_{s=1}^{20,000} (D_s - \bar{D})^2 \right\}^{1/2}. \quad (\text{B.8})$$

Calculating D_s from each sampled pair accounts for the two methods being applied to the same data. Individual-method intervals use Wilson’s binomial score construction (Wilson, 1927). Both types of interval describe Monte Carlo uncertainty for one setting, not simultaneous uncertainty across all settings.

B.5 Validity and sparse-sample diagnostics

Both tests require matching table shapes with at least two rows and columns, finite nonnegative integer counts, and more than one observation in each table. Normal Wald requires a finite corrected MI difference, a positive finite standard error, $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$, and a finite p-value. The variance threshold treats values extremely close to zero as numerically unusable.

Expanded Welch adds component-level conditions to those shared requirements. It requires positive finite degrees of freedom for both populations and their combination. If one component’s degrees of freedom cannot be calculated, the expanded test is invalid even when the direct combined expression in Section A.5 has a value. Failed calculations contribute no rejection to the unconditional rate.

For population P with sample size n_P , the minimum expected cell count is $\min_{ij} n_{Pp_{ij}}$; define Q ’s analogously. Sparsity diagnostics include empty observed rows and columns and the fraction of zero-count cells. They help describe why a statistic might fail but are not used to exclude settings from the study.

B.6 Independent-pilot standard-error check

The diagnostic in Section 6.5 uses two independent sets of 20,000 sampled table pairs from the stated null populations. The pilot set gives the sample standard deviation of the corrected MI differences. The evaluation set is tested twice: once with each pair’s plug-in standard error, and once with that fixed pilot standard deviation. Both calculations use the standard normal distribution and the two-sided 5% threshold. The comparison of variance estimates divides the evaluation set’s average squared plug-in standard error by the squared pilot standard deviation. This check retains the population pair and estimated differences while changing how they are standardised.

Appendix C

Additional Results

These results extend the comparisons in Chapter 6. Each setting uses 20,000 paired samples at the two-sided 5% threshold.

C.1 Individual main-grid null results

Table C.1 gives every main-grid null setting separately, across four table sizes, three margin profiles and nine sample sizes. All have $I(P) = I(Q) = 0.02$ nats. Uniform and same-skew settings have $P = Q$; different-skew settings have $P \neq Q$ with equal MI. The margins are specified in Table 5.1. Rejection rates divide valid rejections by all 20,000 samples; the valid-result rate shows the fraction returning a usable p-value. At $n = 2$, both methods have zero validity. Their zero rejection rates therefore indicate failed calculations, not well-calibrated tests.

Table C.1: False-positive and valid-result rates for all 108 main null settings.

Shape	Margins	n	Normal Wald		Expanded Welch	
			Reject	Valid	Reject	Valid
2×2	Uniform	2	0.0000	0.0000	0.0000	0.0000
2×2	Uniform	5	0.0914	0.9862	0.0000	0.7774
2×2	Uniform	10	0.0588	0.9889	0.0263	0.9795
2×2	Uniform	20	0.0261	0.9977	0.0164	0.9833
2×2	Uniform	50	0.0082	0.9999	0.0045	0.9938
2×2	Uniform	100	0.0142	1.0000	0.0063	0.9991
2×2	Uniform	250	0.0306	1.0000	0.0164	1.0000
2×2	Uniform	500	0.0408	1.0000	0.0298	1.0000

Shape	Margins	n	Normal Wald		Expanded Welch	
			Reject	Valid	Reject	Valid
2×2	Uniform	1000	0.0445	1.0000	0.0386	1.0000
2×2	Same skew	2	0.0000	0.0000	0.0000	0.0000
2×2	Same skew	5	0.1110	0.7282	0.0000	0.2304
2×2	Same skew	10	0.0357	0.9585	0.0092	0.6511
2×2	Same skew	20	0.0232	0.9980	0.0135	0.9550
2×2	Same skew	50	0.0086	1.0000	0.0037	1.0000
2×2	Same skew	100	0.0107	1.0000	0.0047	1.0000
2×2	Same skew	250	0.0260	1.0000	0.0135	1.0000
2×2	Same skew	500	0.0379	1.0000	0.0268	1.0000
2×2	Same skew	1000	0.0437	1.0000	0.0372	1.0000
2×2	Different skew	2	0.0000	0.0000	0.0000	0.0000
2×2	Different skew	5	0.1008	0.8454	0.0000	0.3423
2×2	Different skew	10	0.0432	0.9791	0.0120	0.7575
2×2	Different skew	20	0.0244	0.9983	0.0151	0.9749
2×2	Different skew	50	0.0080	1.0000	0.0039	0.9999
2×2	Different skew	100	0.0121	1.0000	0.0054	1.0000
2×2	Different skew	250	0.0290	1.0000	0.0158	1.0000
2×2	Different skew	500	0.0387	1.0000	0.0283	1.0000
2×2	Different skew	1000	0.0453	1.0000	0.0380	1.0000
3×3	Uniform	2	0.0000	0.0000	0.0000	0.0000
3×3	Uniform	5	0.1931	0.9994	0.0000	0.9534
3×3	Uniform	10	0.1058	1.0000	0.0285	0.9981
3×3	Uniform	20	0.0730	1.0000	0.0343	1.0000
3×3	Uniform	50	0.0288	1.0000	0.0173	1.0000
3×3	Uniform	100	0.0167	1.0000	0.0106	1.0000
3×3	Uniform	250	0.0269	1.0000	0.0178	1.0000
3×3	Uniform	500	0.0355	1.0000	0.0278	1.0000
3×3	Uniform	1000	0.0445	1.0000	0.0393	1.0000
3×3	Same skew	2	0.0000	0.0000	0.0000	0.0000
3×3	Same skew	5	0.1807	0.7217	0.0000	0.2209
3×3	Same skew	10	0.0602	0.9599	0.0176	0.6627

Shape	Margins	n	Normal Wald		Expanded Welch	
			Reject	Valid	Reject	Valid
3×3	Same skew	20	0.0333	0.9996	0.0207	0.9565
3×3	Same skew	50	0.0135	1.0000	0.0076	1.0000
3×3	Same skew	100	0.0139	1.0000	0.0068	1.0000
3×3	Same skew	250	0.0273	1.0000	0.0149	1.0000
3×3	Same skew	500	0.0348	1.0000	0.0235	1.0000
3×3	Same skew	1000	0.0391	1.0000	0.0314	1.0000
3×3	Different skew	2	0.0000	0.0000	0.0000	0.0000
3×3	Different skew	5	0.2089	0.8383	0.0000	0.3296
3×3	Different skew	10	0.0886	0.9886	0.0260	0.7792
3×3	Different skew	20	0.0466	0.9998	0.0251	0.9745
3×3	Different skew	50	0.0188	1.0000	0.0098	1.0000
3×3	Different skew	100	0.0171	1.0000	0.0083	1.0000
3×3	Different skew	250	0.0268	1.0000	0.0147	1.0000
3×3	Different skew	500	0.0331	1.0000	0.0232	1.0000
3×3	Different skew	1000	0.0411	1.0000	0.0348	1.0000
5×5	Uniform	2	0.0000	0.0000	0.0000	0.0000
5×5	Uniform	5	0.2287	1.0000	0.0000	0.9909
5×5	Uniform	10	0.1061	1.0000	0.0665	0.9997
5×5	Uniform	20	0.0768	1.0000	0.0528	1.0000
5×5	Uniform	50	0.0643	1.0000	0.0467	1.0000
5×5	Uniform	100	0.0377	1.0000	0.0289	1.0000
5×5	Uniform	250	0.0232	1.0000	0.0190	1.0000
5×5	Uniform	500	0.0276	1.0000	0.0240	1.0000
5×5	Uniform	1000	0.0326	1.0000	0.0296	1.0000
5×5	Same skew	2	0.0000	0.0000	0.0000	0.0000
5×5	Same skew	5	0.2175	0.7137	0.0000	0.2149
5×5	Same skew	10	0.0825	0.9599	0.0363	0.6756
5×5	Same skew	20	0.0651	0.9996	0.0454	0.9553
5×5	Same skew	50	0.0290	1.0000	0.0178	1.0000
5×5	Same skew	100	0.0112	1.0000	0.0066	1.0000
5×5	Same skew	250	0.0152	1.0000	0.0094	1.0000

Shape	Margins	n	Normal Wald		Expanded Welch	
			Reject	Valid	Reject	Valid
5×5	Same skew	500	0.0369	1.0000	0.0260	1.0000
5×5	Same skew	1000	0.0403	1.0000	0.0329	1.0000
5×5	Different skew	2	0.0000	0.0000	0.0000	0.0000
5×5	Different skew	5	0.2690	0.8356	0.0000	0.3270
5×5	Different skew	10	0.1514	0.9889	0.0693	0.7817
5×5	Different skew	20	0.1259	1.0000	0.0839	0.9785
5×5	Different skew	50	0.0623	1.0000	0.0348	1.0000
5×5	Different skew	100	0.0250	1.0000	0.0136	1.0000
5×5	Different skew	250	0.0254	1.0000	0.0162	1.0000
5×5	Different skew	500	0.0360	1.0000	0.0268	1.0000
5×5	Different skew	1000	0.0386	1.0000	0.0319	1.0000
8×8	Uniform	2	0.0000	0.0000	0.0000	0.0000
8×8	Uniform	5	0.2692	0.9983	0.0000	0.9163
8×8	Uniform	10	0.1376	1.0000	0.0762	0.9996
8×8	Uniform	20	0.0687	1.0000	0.0570	1.0000
8×8	Uniform	50	0.0703	1.0000	0.0592	1.0000
8×8	Uniform	100	0.0712	1.0000	0.0618	1.0000
8×8	Uniform	250	0.0317	1.0000	0.0286	1.0000
8×8	Uniform	500	0.0212	1.0000	0.0198	1.0000
8×8	Uniform	1000	0.0228	1.0000	0.0211	1.0000
8×8	Same skew	2	0.0000	0.0000	0.0000	0.0000
8×8	Same skew	5	0.2266	0.6984	0.0000	0.2082
8×8	Same skew	10	0.0873	0.9587	0.0456	0.6796
8×8	Same skew	20	0.0873	0.9992	0.0692	0.9541
8×8	Same skew	50	0.0537	1.0000	0.0367	1.0000
8×8	Same skew	100	0.0245	1.0000	0.0155	1.0000
8×8	Same skew	250	0.0106	1.0000	0.0071	1.0000
8×8	Same skew	500	0.0162	1.0000	0.0129	1.0000
8×8	Same skew	1000	0.0341	1.0000	0.0289	1.0000
8×8	Different skew	2	0.0000	0.0000	0.0000	0.0000
8×8	Different skew	5	0.2986	0.8319	0.0000	0.3157

Shape	Margins	n	Normal Wald		Expanded Welch	
			Reject	Valid	Reject	Valid
8×8	Different skew	10	0.1853	0.9897	0.1035	0.7912
8×8	Different skew	20	0.1803	1.0000	0.1432	0.9759
8×8	Different skew	50	0.1391	1.0000	0.1080	1.0000
8×8	Different skew	100	0.0921	1.0000	0.0699	1.0000
8×8	Different skew	250	0.0583	1.0000	0.0449	1.0000
8×8	Different skew	500	0.0377	1.0000	0.0300	1.0000
8×8	Different skew	1000	0.0356	1.0000	0.0301	1.0000

C.2 Selected additional square-table curves

The following curves add the intermediate 3×3 and 5×5 shapes to the 2×2 and 8×8 comparisons in Chapter 6. The first uses uniform margins; the second compares different margins. Upper panels use a 0–1 rejection scale and lower panels enlarge 0–0.15. Hollow markers indicate validity below 90%.

For these 3×3 tables, both P and Q have uniform row and column margins $(1/3, 1/3, 1/3)$. From left to right, the columns use equal sample sizes $n_P = n_Q = 10, 100, 1000$. Both populations use probability transfers within the first two rows and columns, with $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The MI difference varies over $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

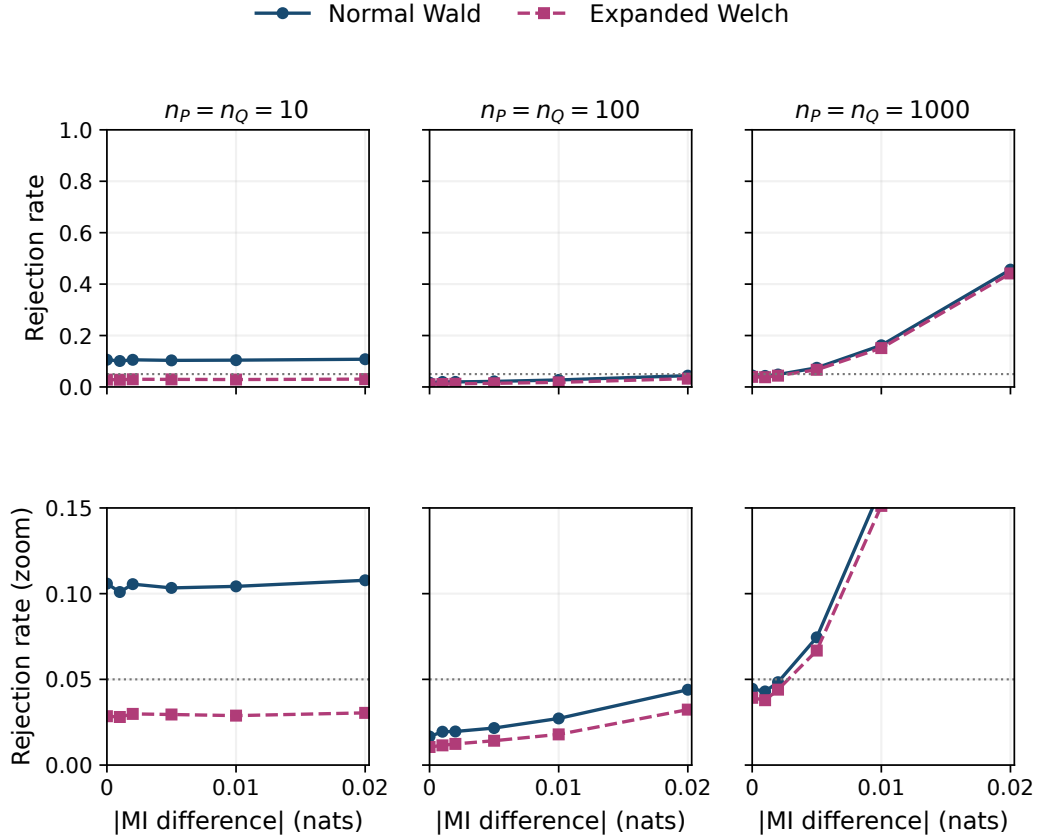


Figure C.1: Normal Wald and Expanded Welch rejection rates for 3×3 tables under the uniform margin profile. The lower panels enlarge the 0–0.15 range.

At $n = 10$, Wald’s zero-difference rate is above 0.05 and Expanded Welch brings it below the line; both methods are almost always valid. At $n = 100$, however, both start well below 0.05 and the correction lowers rejection further. Detection grows strongly at $n = 1,000$, where Wald remains slightly higher. Whether the adjustment helps calibration changes with sample size, not just table shape.

For these 5×5 tables, the margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$ for P and $(0.8, 0.05, 0.05, 0.05, 0.05)$ for Q , each used for both rows and columns. From left to right, the columns use equal sample sizes $n_P = n_Q = 10, 100, 1000$. Both populations use probability transfers within the first two rows and columns, with $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The MI difference varies over $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

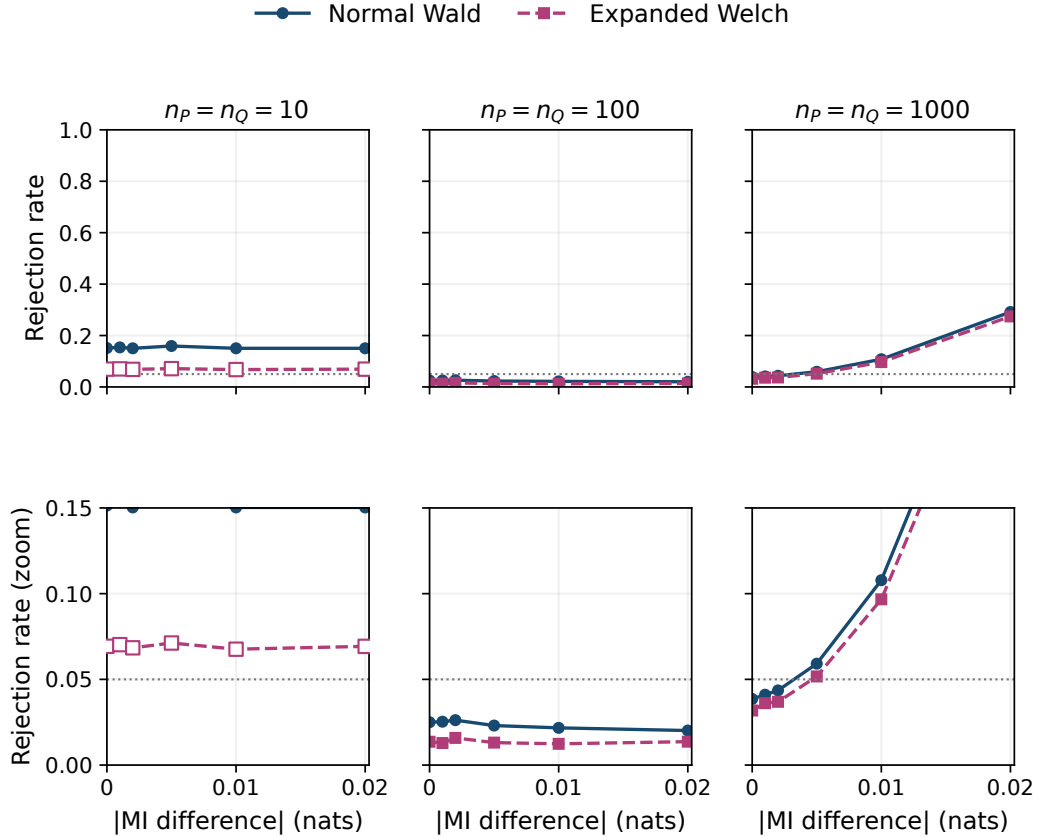


Figure C.2: Normal Wald and Expanded Welch rejection rates for 5×5 tables under the different skew margin profile. The lower panels enlarge the 0–0.15 range.

The $n = 10$ null is liberal for both methods, though less so for Expanded Welch; its hollow markers also show reduced validity. At $n = 100$, rejection stays low and falls across this short alternative range. At $n = 1,000$, detection rises and Wald remains slightly higher. Expanded Welch lowers rejection at every size, helping only where the null rate is too high.

C.3 Other controlled comparisons

The following comparisons change association location, sample allocation, table shape or population construction. The final figure starts at exact independence, where the first-order MI variance is zero.

Figure C.3 compares the location of association in 5×5 tables. Columns place association in the first 2×2 block, last (rare) 2×2 block, or a centred score outer product spread across all cells. For every column, P 's row and column margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$, and Q 's are

$(0.8, 0.05, 0.05, 0.05, 0.05)$. Here $I(P) = 0.0001$, $I(Q) = I(P) + |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.0001, 0.00025, 0.0005, 0.001, 0.002$ nats.

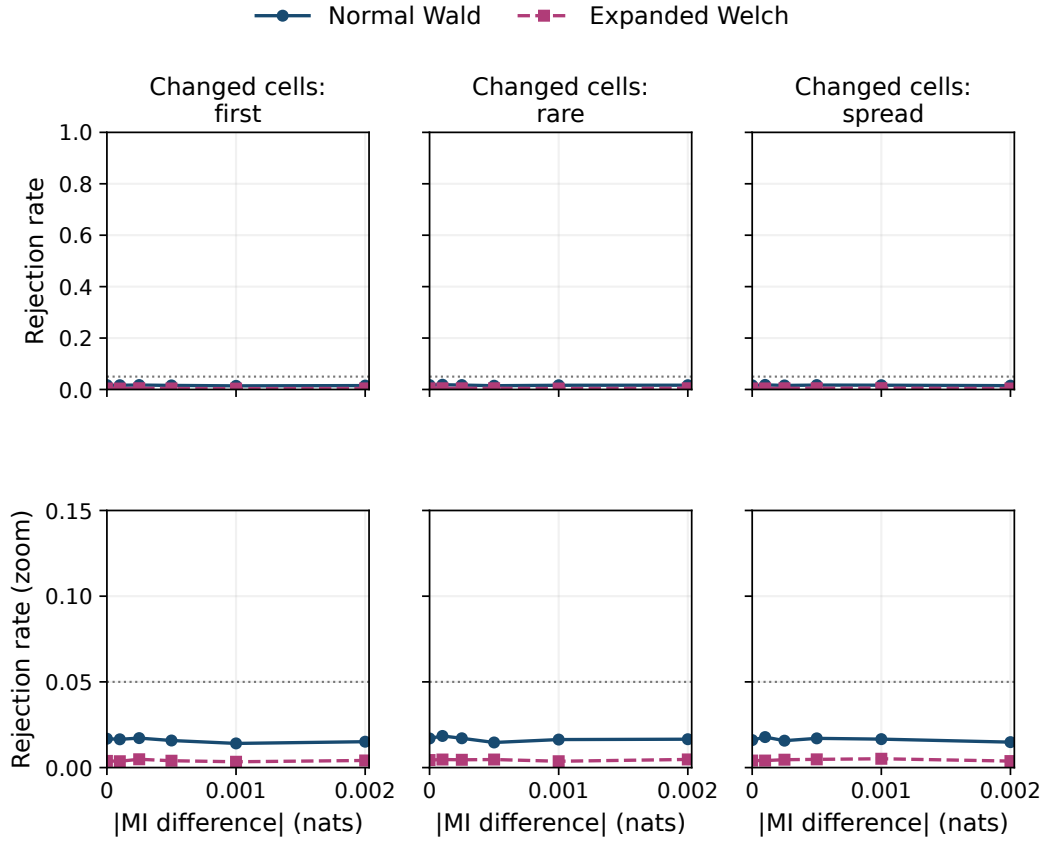


Figure C.3: Rejection rates for 5×5 tables with association in different cells.

All three association patterns give below-target null rejection in this comparison. For first, rare and spread patterns, respectively, the zero-difference Wald rates are 0.0169, 0.0170 and 0.0160; Expanded Welch gives 0.0039, 0.0044 and 0.0041. All displayed calculations are valid. The alternative curves remain nearly flat over this short MI range, so neither method detects these differences well. The low null rejection and limited detection persist across the three association patterns.

Figure C.4 compares unequal samples when Q has the higher MI. Columns have $(n_P, n_Q) = (50, 250)$ and $(250, 50)$. For both columns, P has 3×3 row and column margins $(0.7, 0.15, 0.15)$ and Q has $(0.8, 0.1, 0.1)$; the first-cell change block is used. $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

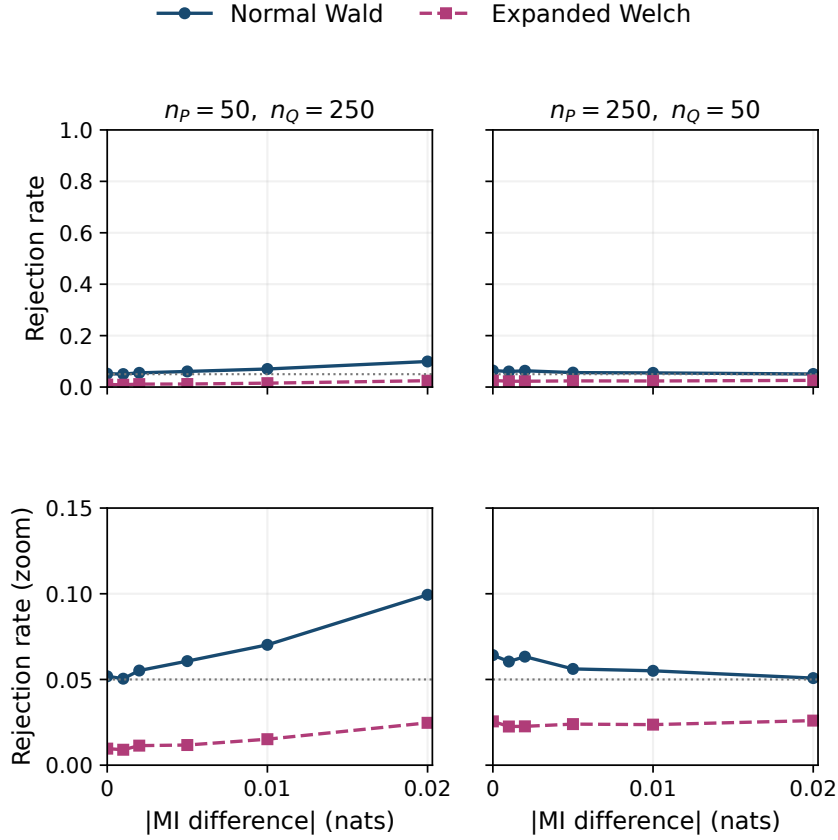


Figure C.4: Rejection rates under two sample allocations when Q has the higher MI.

Allocating more observations to Q , the population with higher MI, gives stronger detection in this comparison. At zero difference, Wald is near or above 0.05 while Expanded Welch is below it in both allocations. When Q has more observations $(n_P, n_Q) = (50, 250)$, rejection rises with $I(Q) - I(P)$; at 0.02 nats it reaches 0.0994 for Wald and 0.0247 for Expanded Welch. With P larger, the curves rise much less. These points are valid, but the alternative gap is not a comparison at matched false-positive rates: the null rates differ even though both tests use the nominal 5% threshold.

Figure C.5 reverses the MI direction. Columns have $(n_P, n_Q) = (50, 250)$ and $(250, 50)$. P 's 3×3 row and column margins are $(0.7, 0.15, 0.15)$; Q 's are $(0.8, 0.1, 0.1)$. Both use the first-cell change block. $I(Q) = 0.02$ and $I(P) = 0.02 + |\Delta|$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

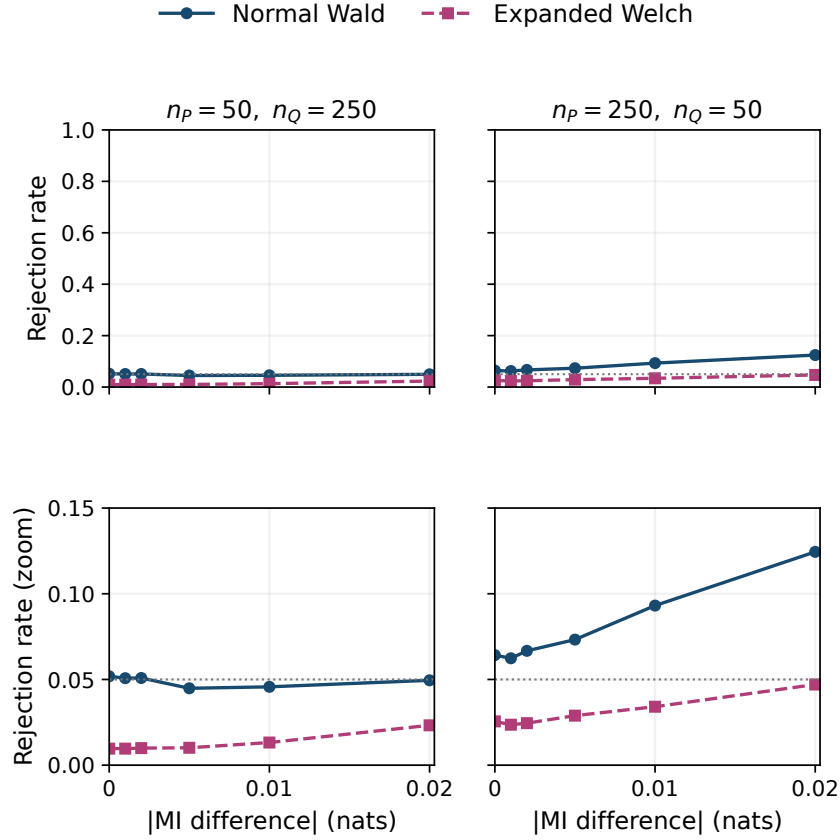


Figure C.5: Rejection rates under two sample allocations when P has the higher MI.

Reversing the MI direction also reverses which sample allocation gives stronger detection. The zero-difference rates are the same as in Figure C.4, because each allocation uses the same pair of populations in both figures at that point; P and Q still have different margins. Reversing the MI direction changes what happens next: when P has the larger sample, Wald rises well above its null rate, whereas it stays nearly flat when Q has the larger sample. At $(n_P, n_Q) = (50, 250)$ and 0.02 nats, Wald rejects 0.0495 and Expanded Welch 0.0232; both calculations are valid. Direction and allocation therefore matter even at the same absolute MI difference.

Figure C.6 compares rectangular tables. Columns are 2×3 and 3×5 . In each dimension k , P 's corresponding margin is $m_k(0.7)$ and Q 's is $m_k(0.8)$, for both rows and columns. Thus the first column uses P rows $(0.7, 0.3)$, columns $(0.7, 0.15, 0.15)$, and Q rows $(0.8, 0.2)$, columns $(0.8, 0.1, 0.1)$; the second applies the same formula to 3 and 5 categories. Both use first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.02$, and $I(Q) = 0.02 + |\Delta|$ for $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

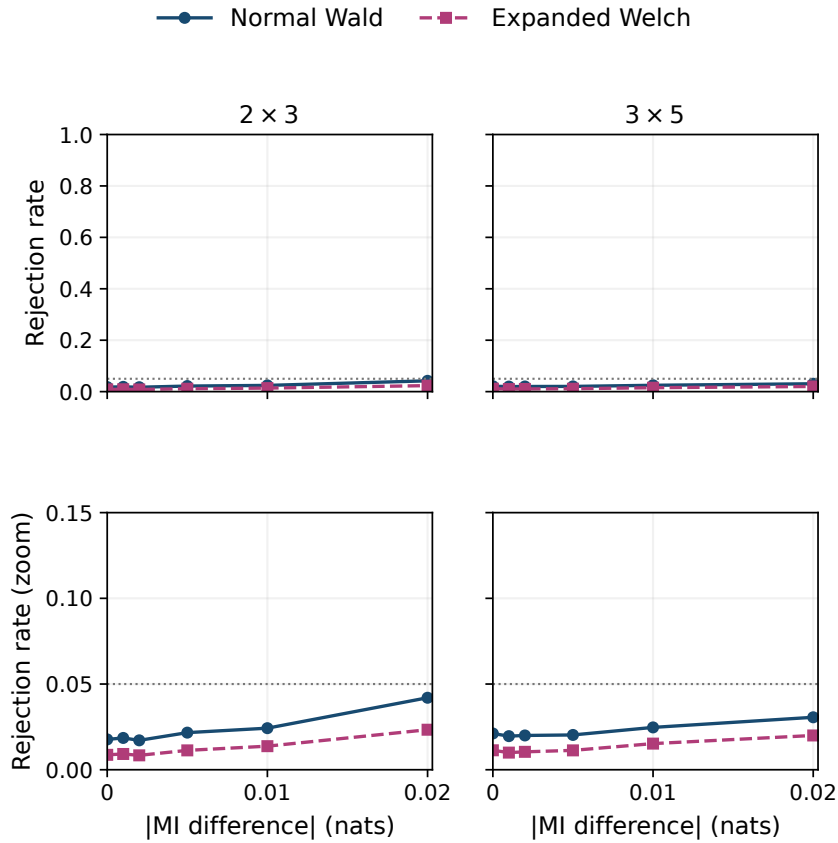


Figure C.6: Rejection rates for 2×3 and 3×5 tables with different margins.

Both rectangular panels start below 0.05, with Expanded Welch lower. Rejection grows only modestly by 0.02 nats and remains below 0.05 for both methods throughout the displayed range; validity is high. Both methods therefore detect these differences poorly. Expanded Welch also lowers an already-conservative null rate, as in the square tables.

Figure C.7 compares additive and ordinal log-linear constructions for 5×5 tables. The left column uses the first-cell additive block; the right fits an ordinal interaction while preserving the margins. In both, P 's row and column margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$ and Q 's are $(0.8, 0.05, 0.05, 0.05, 0.05)$. Both have $I(P) = 0.02$, $I(Q) = 0.02 + |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

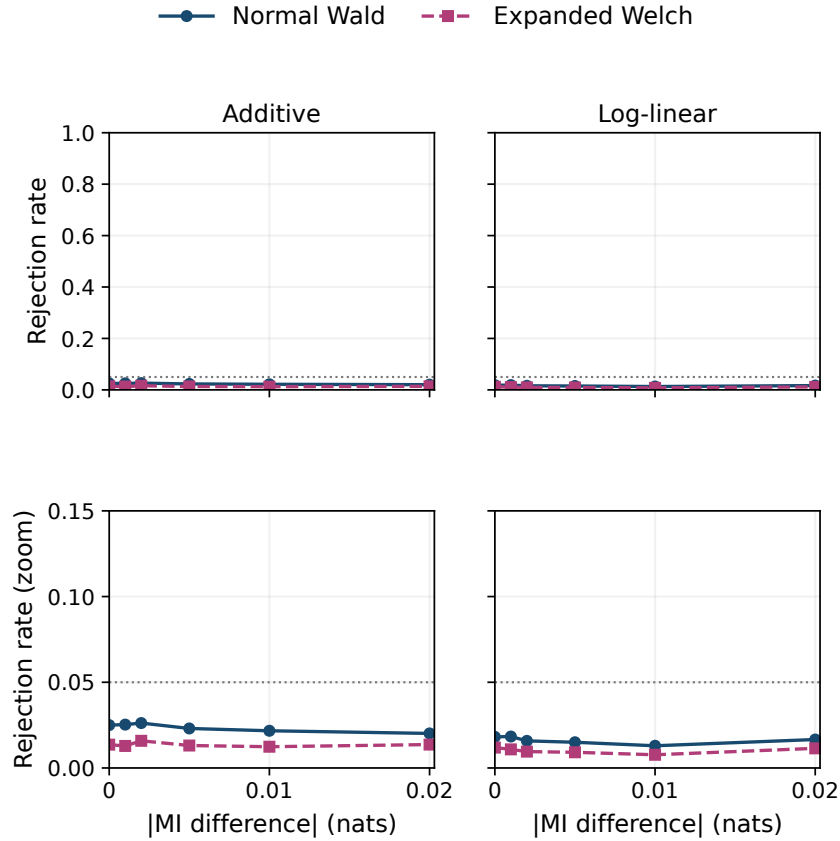


Figure C.7: Rejection rates under additive and ordinal log-linear constructions for 5×5 tables.

Both constructions give conservative null rejection despite producing different cell probabilities. At zero difference, the additive and log-linear Wald rates are 0.0250 and 0.0181; Expanded Welch gives 0.0136 and 0.0119. All four are valid and below 0.05. Rejection stays low or falls slightly as the displayed difference grows. Matching margins and MI therefore does not ensure identical rejection rates.

Figure C.8 starts at exact independence in 2×2 tables. Columns are uniform margins $(0.5, 0.5)$ for both populations, same skew $(0.8, 0.2)$ for both, and different skew with $P = (0.7, 0.3)$ and $Q = (0.8, 0.2)$; each vector is used for both rows and columns. The first-cell block is used, $I(P) = 0$, $I(Q) = |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats.

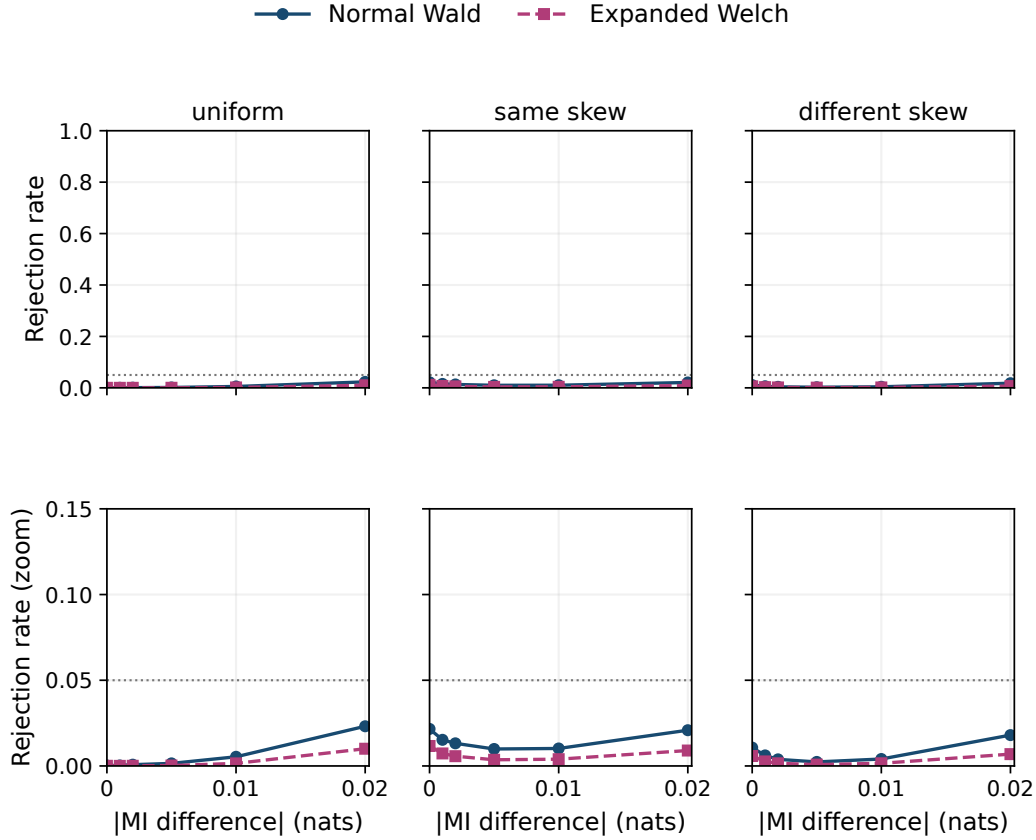


Figure C.8: Rejection rates for 2×2 tables starting at exact independence.

At the uniform zero-difference point, Wald rejects 0.00025 and Expanded Welch 0.00010 of samples; validity is high. Rejection grows little over most of the displayed alternative range, and Expanded Welch remains lower. At exact independence, the population first-order MI variance is zero, so the approximation underlying both curves changes. Section A.4 explains the different reference used by a one-table independence test.

Bibliography

- Robert M. Bell and Daniel F. McCaffrey. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181, 2002. URL <https://www150.statcan.gc.ca/n1/pub/12-001-x/2002002/article/9058-eng.pdf>.
- Thomas B. Berrett and Richard J. Samworth. USP: An independence test that improves on Pearson’s chi-squared and the G -test. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 477(2256):20210549, 2021. doi: 10.1098/rspa.2021.0549.
- Terry Bossomaier, Lionel Barnett, Michael Harré, and Joseph T. Lizier. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer, Cham, 2016. doi: 10.1007/978-3-319-43222-9.
- Richard P. Brent. *Algorithms for Minimization without Derivatives*. Prentice-Hall, Englewood Cliffs, New Jersey, 1973.
- David R. Brillinger. Some data analyses using mutual information. *Brazilian Journal of Probability and Statistics*, 18(2):163–183, 2004. URL <https://www.stat.berkeley.edu/~brill/Papers/MIBJPS.pdf>.
- EunYi Chung and Joseph P. Romano. Exact and asymptotically robust permutation tests. *Annals of Statistics*, 41(2):484–507, 2013. doi: 10.1214/13-AOS1090.
- W. Edwards Deming and Frederick F. Stephan. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics*, 11(4):427–444, 1940. doi: 10.1214/aoms/1177731829.
- Robert Drake and Apratim Guha. A mutual information-based k -sample test for discrete distributions. *Journal of Applied Statistics*, 41(9):2011–2027, 2014. doi: 10.1080/02664763.2014.899325.
- Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.

- Michael P. Fay and Barry I. Graubard. Small-sample adjustments for Wald-type tests using sandwich estimators. *Biometrics*, 57(4):1198–1206, 2001. doi: 10.1111/j.0006-341X.2001.01198.x.
- Apratim Guha and Tom Chothia. A two sample test based on mutual information. *Calcutta Statistical Association Bulletin*, 66(1–2):39–54, 2014. doi: 10.1177/0008068320140103.
- Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974. doi: 10.1080/01621459.1974.10482962.
- Kermit Hutcheson. A test for comparing diversities based on the Shannon formula. *Journal of Theoretical Biology*, 29(1):151–154, 1970. doi: 10.1016/0022-5193(70)90124-4.
- Marcus Hutter. Distribution of mutual information. In *Advances in Neural Information Processing Systems 14*, pages 399–406, 2002. Conference held in 2001; arXiv:cs/0112019.
- Marcus Hutter and Marco Zaffalon. Distribution of mutual information from complete and incomplete data. *Computational Statistics & Data Analysis*, 48(3): 633–657, 2005. doi: 10.1016/j.csda.2004.03.010.
- Guido W. Imbens and Michal Kolesár. Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*, 98(4):701–712, 2016. doi: 10.1162/REST_a_00552.
- Arnold Janssen. Studentized permutation tests for non-i.i.d. hypotheses and the generalized Behrens–Fisher problem. *Statistics & Probability Letters*, 36(1):9–21, 1997. doi: 10.1016/S0167-7152(97)00043-6.
- Kirthevasan Kandasamy, Akshay Krishnamurthy, Barnabas Póczos, Larry Wasserman, and James M. Robins. Nonparametric von Mises estimators for entropies, divergences and mutual informations. In *Advances in Neural Information Processing Systems 28*, pages 397–405, 2015.
- Göran Kauermann and Raymond J. Carroll. A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association*, 96(456):1387–1396, 2001. doi: 10.1198/016214501753382309.
- Marius Marinescu and Costel Balcau. On the use of mutual information for testing independence, 2025. arXiv:2502.17636.

- George A. Miller. Note on the bias of information estimates. In Henry Quastler, editor, *Information Theory in Psychology: Problems and Methods*, pages 95–100. Free Press, Glencoe, Illinois, 1955.
- Roelof Moddemeijer. On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, 16(3):233–248, 1989. doi: 10.1016/0165-1684(89)90132-1.
- Roelof Moddemeijer. A statistic to estimate the variance of the histogram-based mutual information estimator based on dependent pairs of observations. *Signal Processing*, 75(1):51–63, 1999. doi: 10.1016/S0165-1684(98)00224-2.
- Ricardo Mora and Javier Ruiz-Castillo. The statistical properties of the mutual information index of multigroup segregation. Working Paper, Economics Series 48 09-84, Universidad Carlos III de Madrid, Departamento de Economía, 2009.
- Ricardo Mora and Javier Ruiz-Castillo. Entropy-based segregation indices. *Sociological Methodology*, 41(1):159–194, 2011. doi: 10.1111/j.1467-9531.2011.01237.x.
- Wei Pan and Melanie M. Wall. Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21(10):1429–1441, 2002. doi: 10.1002/sim.1142.
- Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272.
- Mark S. Roulston. Estimating the errors on measured entropy and mutual information. *Physica D: Nonlinear Phenomena*, 125(3–4):285–294, 1999. doi: 10.1016/S0167-2789(98)00269-3.
- Franklin E. Satterthwaite. An approximate distribution of estimates of variance components. *Biometrics Bulletin*, 2(6):110–114, 1946. doi: 10.2307/3002019.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948. Parts I and II.
- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.
- Abraham Wald. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54(3):426–482, 1943. doi: 10.1090/S0002-9947-1943-0012401-3.

- Bernard L. Welch. The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947. doi: 10.1093/biomet/34.1-2.28.
- Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938. doi: 10.1214/aoms/1177732360.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.